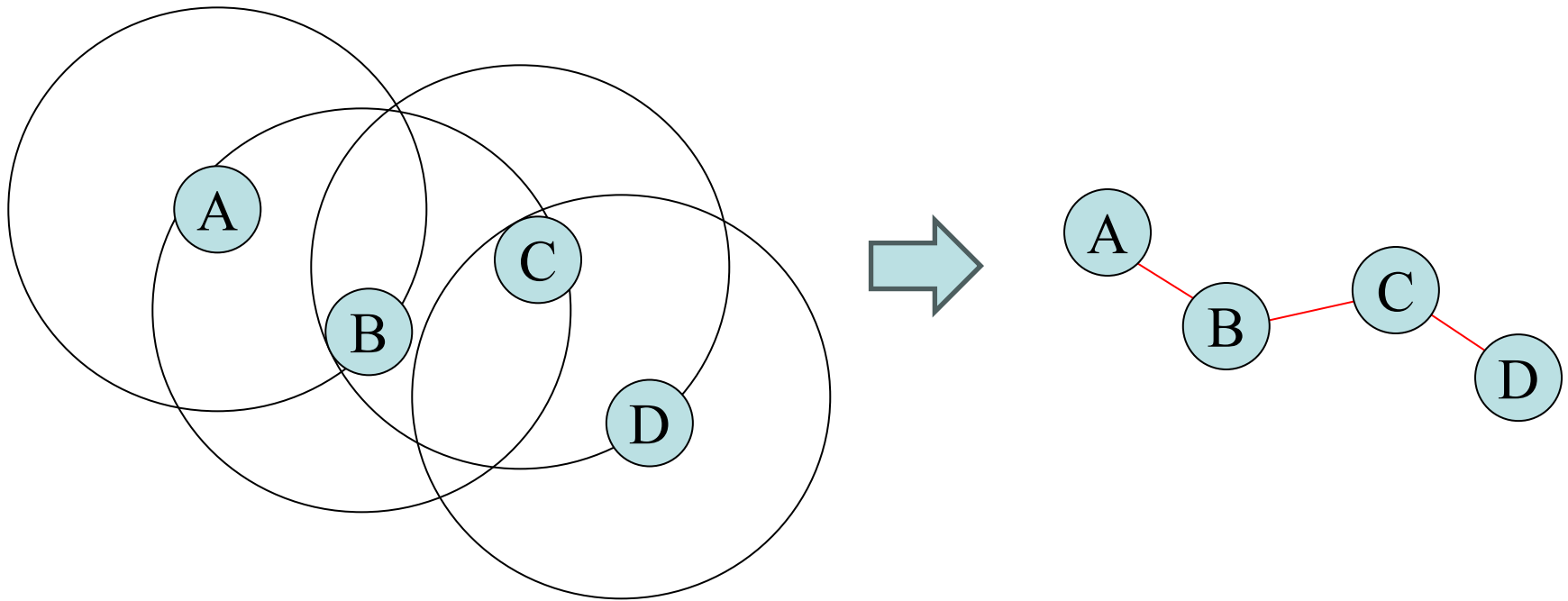


Routing in mobile ad-hoc networks

Multi-Hop Wireless

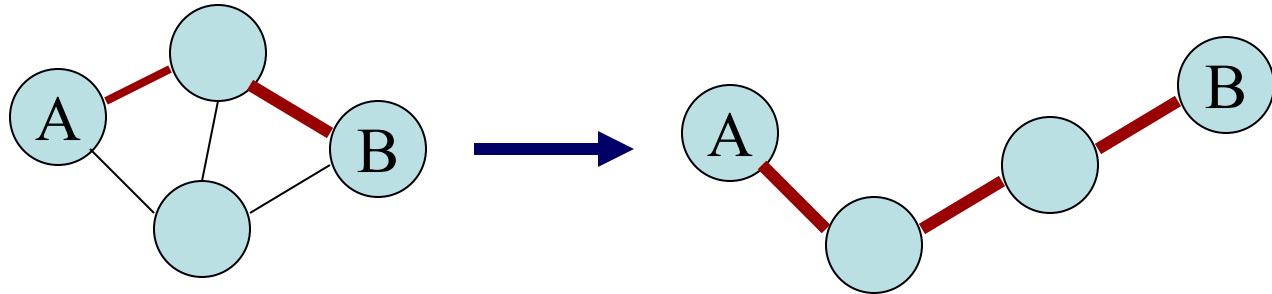
- May need to traverse multiple links to reach destination



- Mobility causes route changes

Mobile Ad Hoc Networks (MANET)

- Host movement frequent
- Topology change frequent



- No cellular infrastructure.
- Multi-hop wireless links.
- Data must be routed via intermediate nodes.

Characteristics

- Each node serves as a router and forwards packets for other nodes in the network.
- Rapidly deployable, self configuring.
- Independent of any fixed infrastructure or centralized administration (no “access point” or backbone).
- Topology can be very dynamic.
- Bandwidth-constrained variable-capacity links
- Limited physical security
- Nodes with limited battery life and storage capabilities

Applications

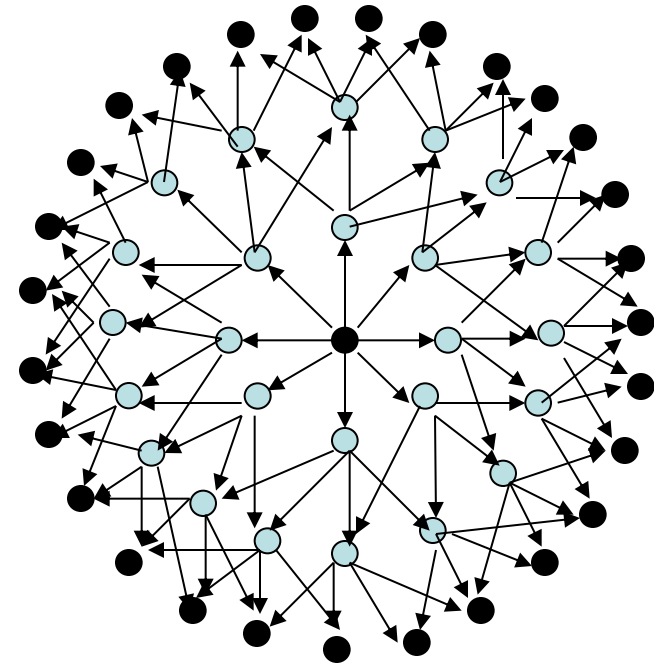
- **Sensor Networks**
 - Ad hoc adaptive field monitoring systems
- **Military**
 - Rapidly deployable battle-site networks
 - Unmanned aerial vehicles, drone swarms
- **Disaster management**
 - Disaster relief teams
 - Rescue Operations
- **Neighborhood area networks (NANs)**
 - Shareable Internet access in high-density urban settings
 - Students on campus
- **Impromptu communications among people**
 - Ubiquitous/pervasive connectivity for meetings
 - Wearable computing
- **Automobile communications**
 - VANETS

Distance-vector & Link-state Routing

- Both assume router knows
 - address of each neighbor
 - cost of reaching each neighbor
- Both allow a router to determine global routing information by talking to its neighbors
- **Distance vector** - router knows cost to each destination
- **Link state** - router knows entire network topology and computes shortest path

Link State Routing effects

- Each node periodically floods status of its links
- Each node re-broadcasts link state information received from its neighbour
- Each node keeps track of link state information received from other nodes
- Each node uses above information to determine next hop to each destination
- Dynamicity introduces extreme protocol overhead

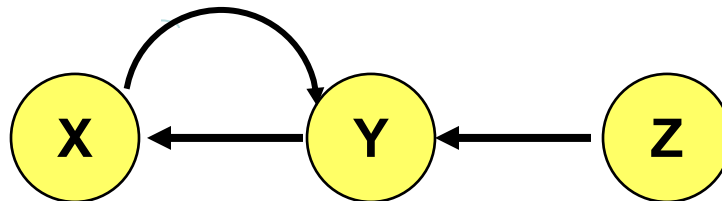


24 retransmissions to diffuse
a message up to 3 hops

● Retransmission node

Distance vector routing effects

- Each node periodically gives neighbors its network view
- Each node, after upgrading its view, rebroadcasts such information to its neighbors
- But here nodes continuously attach and detach and topology changes
- Routing loops arise
- Dynamicity introduces convergence problems



New routing protocol requirements

Why is it different from routing in other types of network? Because both end nodes and routers are mobile

- Self starting and self organizing
- Multi-hop operation with a routing mechanism designed for mobile nodes
- Dynamic topology maintenance
- Rapid convergence
- Minimal network traffic overhead
- Scalable to large networks

MANET Routing Protocols

- **Proactive protocols**

- Traditional distributed shortest-path protocols
- Maintain routes between every host pair at all times
- Based on periodic updates; High routing overhead
- Example: DSDV (destination sequenced distance vector)

- **Reactive protocols**

- Determine route if and when needed
- Source initiates route discovery
- Example: DSR (dynamic source routing)

- **Hybrid protocols**

- Adaptive; Combination of proactive and reactive
- Example : ZRP (zone routing protocol)

Reactive (on-demand) Routing

- **Key Goal:** Reduction in routing overhead
- Useful when number of traffic sessions is much lower than the number of nodes.
- No routing structure created *a priori*. Let the structure emerge in response to a need
- Two key methods for route discovery
 - source routing
 - backward learning (similar to intra-AS routing)
- **Introduces delay**

Reactive (on-demand) routing

- Routing only when needed

Advantages:

- eliminate periodic updates
- adaptive to network dynamics

Disadvantages:

- high flood-search overhead with mobility, distributed traffic
- high route acquisition latency

Reactive Routing – Source initiated

- Source floods the network with a *route request* packet when a route is required to a destination
 - Flood is propagated outwards from the source
 - Pure flooding = every node transmits the request only once
- Destination *replies* to request
 - Reply uses reversed path of route request
 - sets up the forward path
- Two key protocols: DSR and AODV

Protocol Trade-offs

- **Proactive protocols**

- Always maintain routes
- Little or no delay for route determination
- Consume bandwidth to keep routes up-to-date
- Maintain routes which may never be used

- **Reactive protocols**

- Lower overhead since routes are determined on demand
- Significant delay in route determination
- Employ flooding (global search)
- Control traffic may be bursty

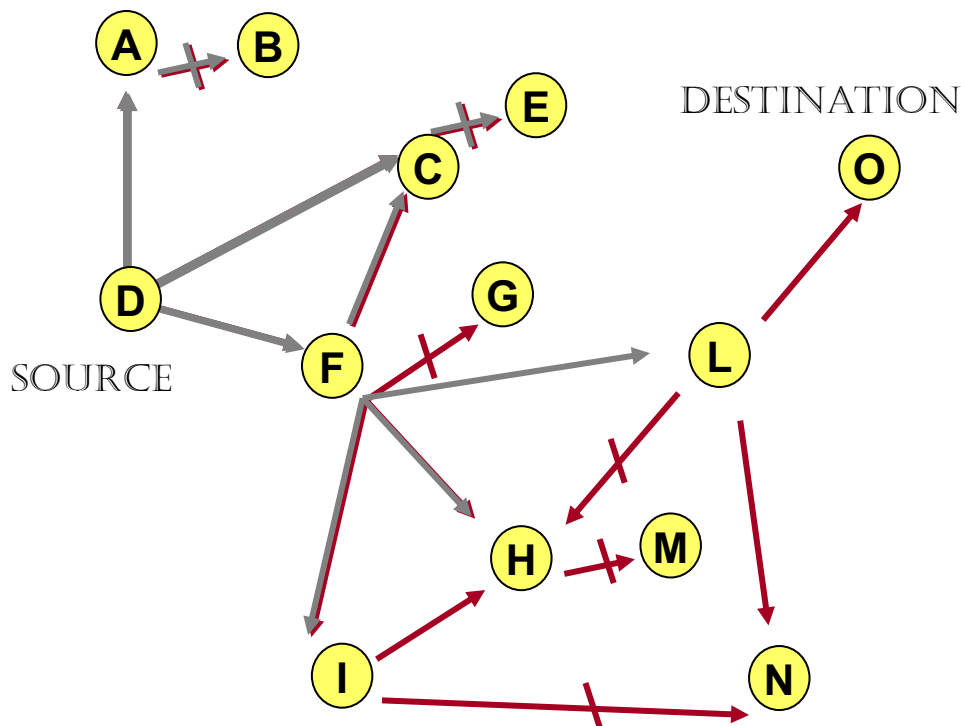
- Which approach achieves a better trade-off depends on the traffic and mobility patterns

Flooding

- A packet is sent by a source node to every one of its neighbors
- At each node, an incoming packet is retransmitted on all outgoing link except for the link on which it arrived
- Hop count field deals with duplicate copies of a pkt
- Properties
 - All possible routes between source and destination are tried
 - At least one copy of the packet to arrive at the destination will have a minimum-hop route
 - All nodes connected to the source node are visited

Gossiping vs. Flooding

Gossip: Probabilistic Flooding



— Gossip-Based Routing

- Node forward packets with some probability $p_G < 1$
- Assign a probability to each outgoing link and to select the link based on that probability

– How good is it?

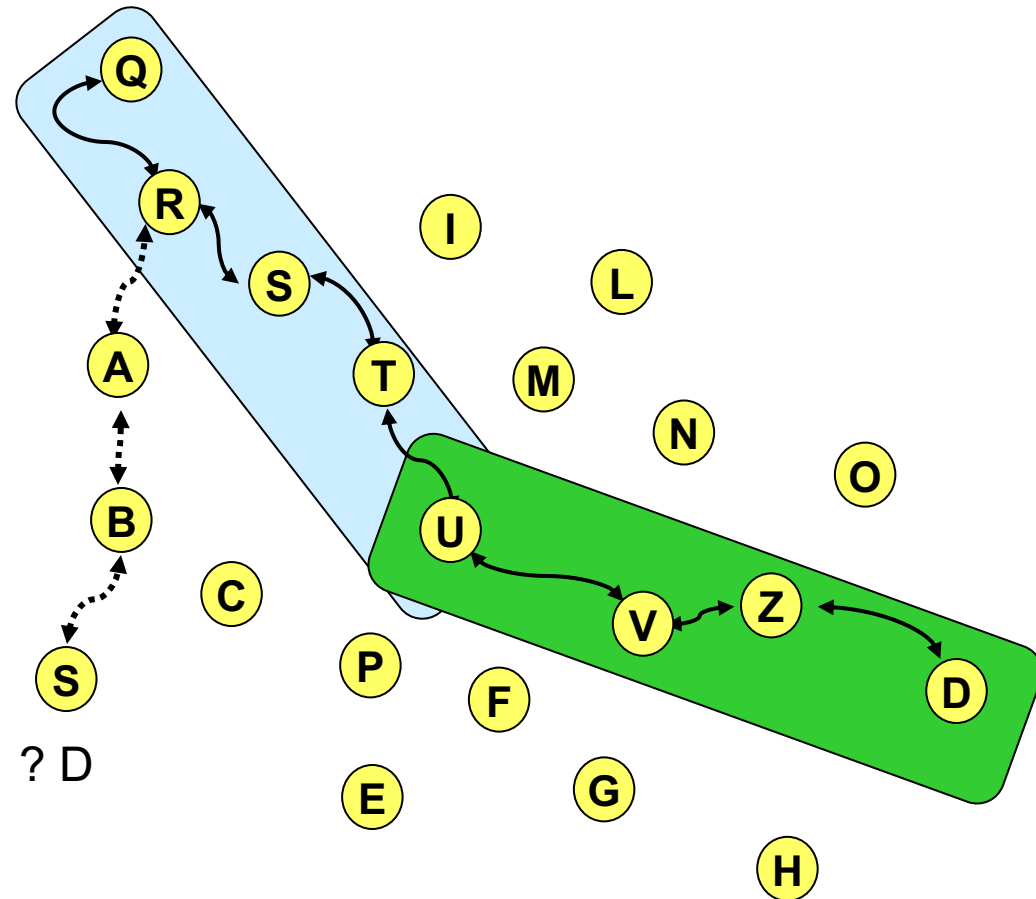
- less overhead than flooding
- Does not guarantee success

– What determines p_G ?

Network Connectivity

Gossiping Agent

- Turn routing problem around: Think of an “agent” wandering through the network, looking for data (events, ...)
- Agent initially perform random walk
- Leave “traces” in the network
- Later agents can use these traces to find data
- Essentially: works due to high probability of line intersections



Connectivity between nodes

- How are the different nodes connected?
- What is the degree of randomness?
- What is the probability distribution function of connectivities of all nodes?
- Interesting properties:
 - Degree and Degree Distribution
 - Maximum Distance between nodes
 - Clustering

Degree and Degree Distribution

- $N_k \equiv$ the number of vertices having $degree(v) = k$ for each $0 < k < v$

- Probability Distribution

$$P(degree(v) = k) = \frac{N_k}{n}$$

$= P(k)$ for short

- Cumulative Distribution

$$P(degree(v) \geq k) = \sum_{i=k}^{\infty} P(i)$$

- Every node has some degree

$$\sum_{i=0}^{\infty} P(i) = 1$$

Connectedness

- **Node** Connectedness:

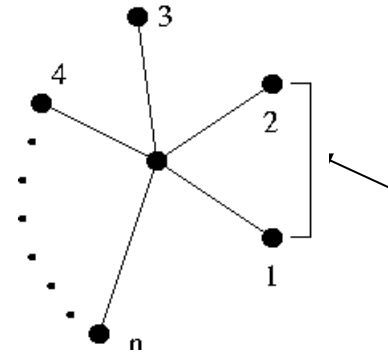
- Nodes i and j are *connected* if there is a path between them.
- They are *disconnected* if such a path does not exist, in which case we have $d_{ij} = \infty$

- **Network** Connectedness:

- A *network is connected* if all pairs of nodes in the network are connected.
- A *network is disconnected* if there is at least one pair with $d_{ij} = \infty$.

Clustering Coefficient

- The clustering coefficient captures the degree to which the neighbors of a given node link to each other
- Characterizes tendency of nodes to cluster
 - Networks are clustered [large C] but have a small characteristic path length [small L].

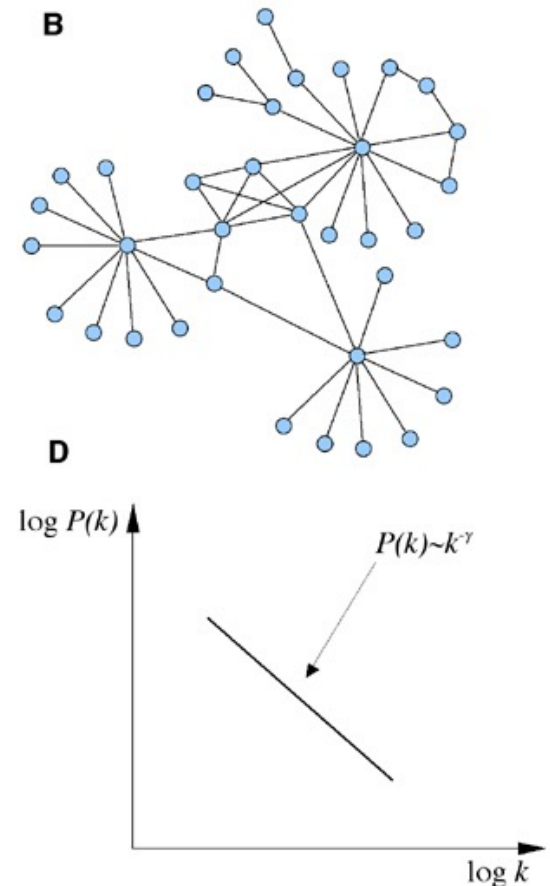


Network	C	C_{rand}	L	N
WWW	0.1078	0.00023	3.1	153127
Internet	0.18-0.3	0.001	3.7-3.76	3015-6209
Actor	0.79	0.00027	3.65	225226
Coauthorship	0.43	0.00018	5.9	52909
Metabolic	0.32	0.026	2.9	282
Foodweb	0.22	0.06	2.43	134
C. elegance	0.28	0.05	2.65	282

The Power Law Distribution

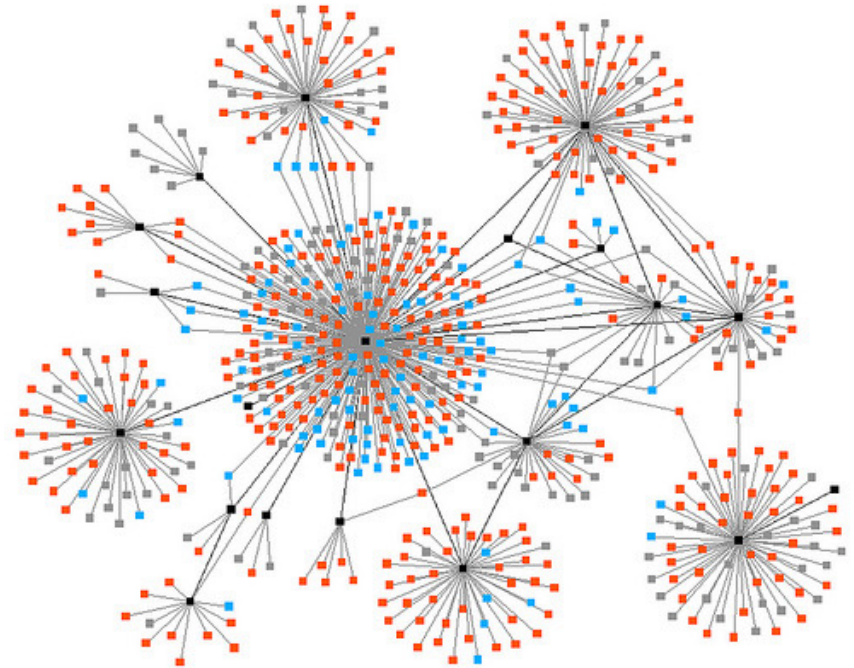
- In general terms, a probability distribution follows a “power law” if:
 - The probability $P(k)$ that a given variable k has a specific value $\alpha k^{-\gamma}$ decreases proportionally to k power $-\gamma$ where γ is a constant value
- For networks, this implies that
 - The probability for a node to have k edges connected is proportional to:

$$P(k) = \alpha k^{-\gamma}$$



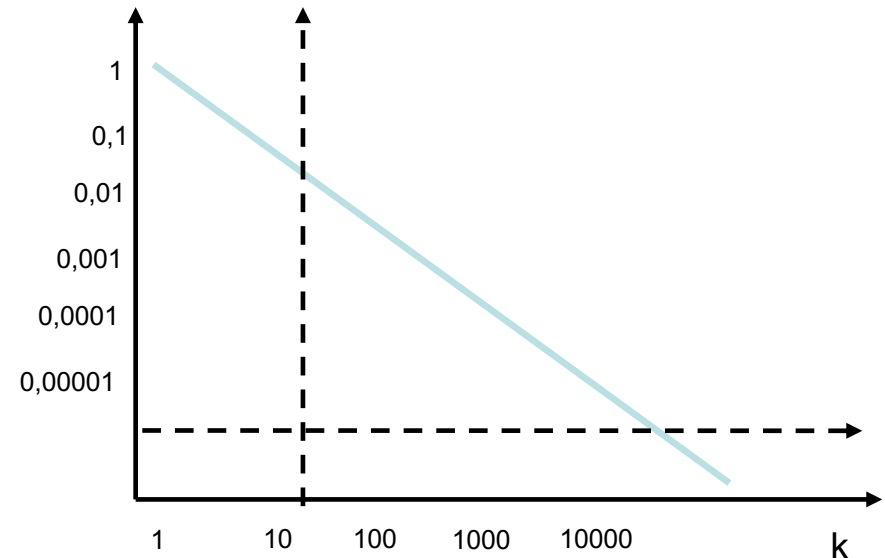
Emergence of Hubs and Connectors

- These networks exhibit the presence of nodes that
 - Act as hubs (assuming high degree), to which most of the other nodes connects to
 - Act as connectors, i.e., nodes that make a great contributions in getting great portion of the network together
 - “smaller nodes” exists that act as hubs or connectors for local portion of the network
 - These hubs serve as the common connections mediating the short path lengths between other edges.



“Scale-Free” Networks

- Networks following a power law distribution for links are called “**scale free**”?
 - Whatever the scale at which we observe the network it looks the same, i.e., it looks similar to itself (i.e. it is **self-similar**)
- The overall properties of the network are preserved independently of the scale:
 - If we cut off the details of a network – e.g. skipping all nodes with a limited number of links – the network will preserve its power-law structure
 - If we consider a sub-portion of any network, it will have the same overall structure of the whole network (due to **self-similarity**)



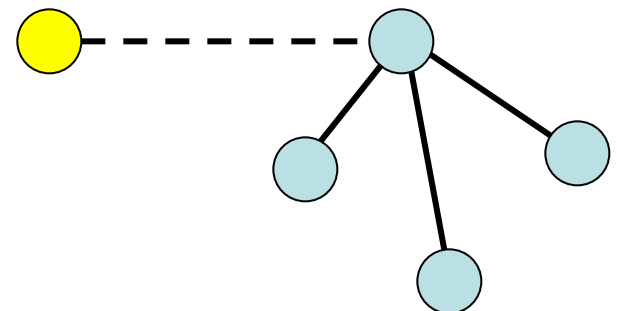
The Power Law dynamics

- Power Law networks are characterized by two fundamental dynamics
 - **Growth**: degree- m nodes are constantly added
 - **Preferential attachment**: the probability that a new node connects to an existing one is proportional to its degree

$$P(k_i) = \frac{k_i}{\sum_j k_j}$$

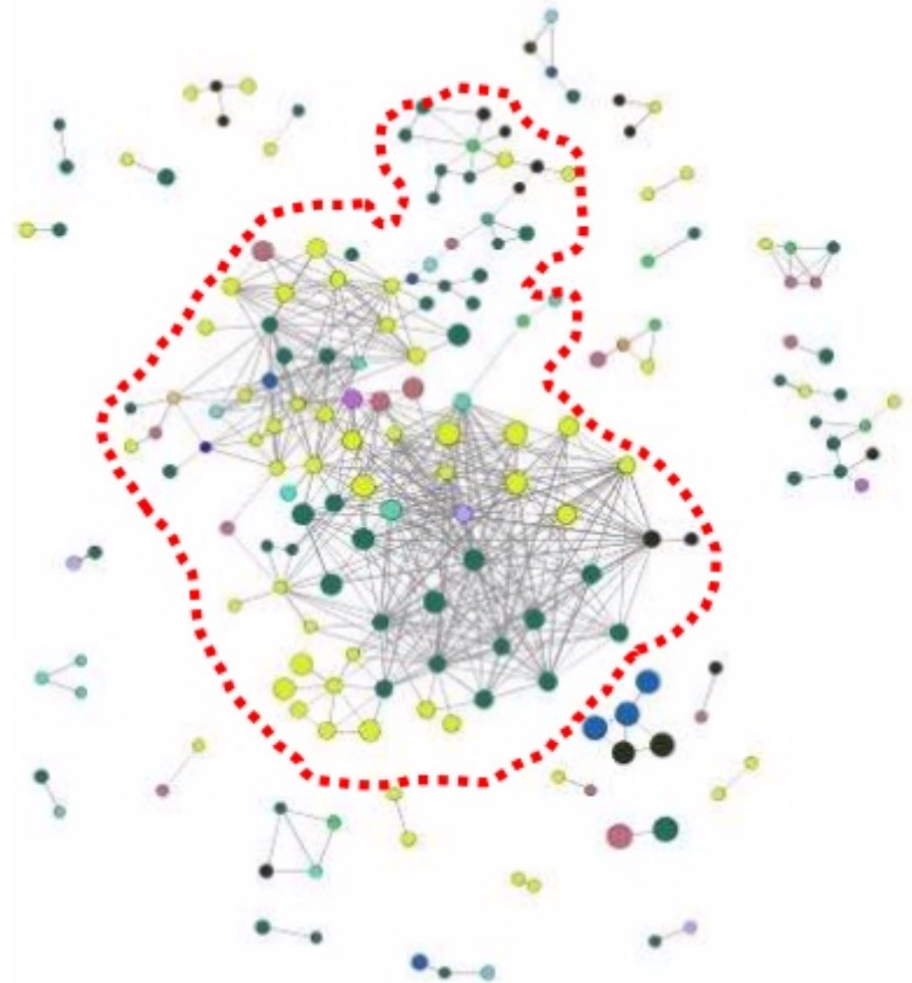
- “The rich get richer” principle emerges

$$P(k) = \frac{2m(m+1)}{(k+2)(k+1)k} \sim k^{-3}$$



Growth dynamics in scale free networks

- Highly connected clusters tend to emerge
- Making a network grow with new nodes that enter the network in successive times and attach preferentially to nodes that already have many links lead to a network structure that is small world and clustered



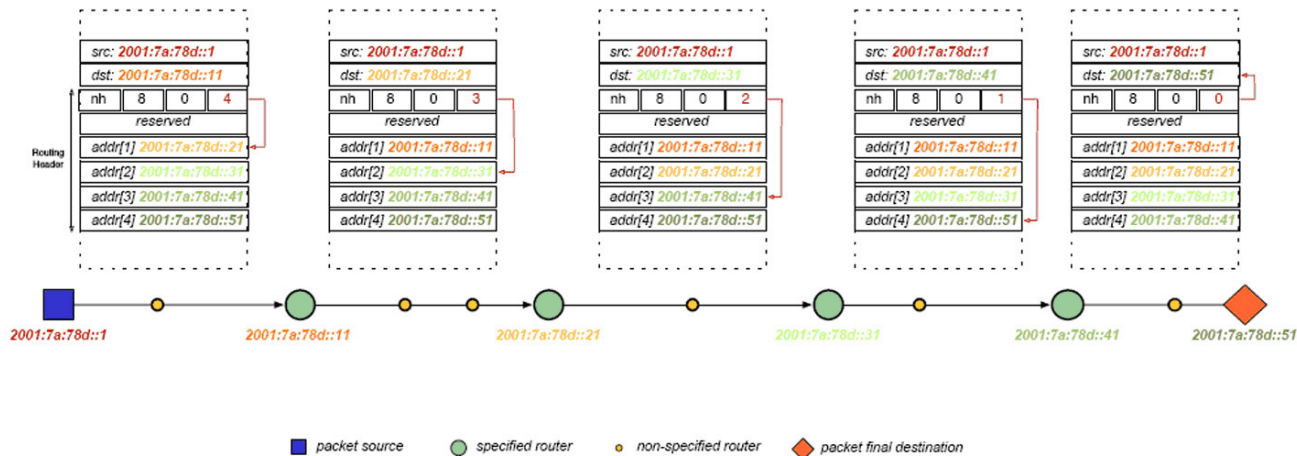
Small World Networks

- Most nodes in real networks are not neighbors, but can be reached from every other one in very few hops.
- The “small world” property implies the existence of a minimum average distance L between nodes
- The distance between two randomly chosen nodes grows proportionally to the logarithm of the number of nodes n in the network, that is:

$$L \propto \log n$$

Source Routing

- In conventional routing, routers in the network determine the path incrementally based on the packet's destination.
- Source routing allows a sender of a packet to partially or completely specify the route the packet takes through the network



- It allows easier troubleshooting, improved traceroute, and enables a node to discover all the possible routes to a host.
- It does not allow a source to directly manage network performance by forcing packets over a path to prevent congestion

Proactive Routing Protocols Examples

Distance vector scheme: DSDV

Link state scheme: OLSR

Destination-Sequenced Distance-Vector (DSDV)

- Each node maintains a routing table which stores
 - **next hop, cost metric** towards each **destination**
 - a **sequence number** that is created by the destination itself
- Each route is **tagged** with the sequence number
 - routes with **greater sequence numbers** are preferred
- Each node periodically forwards routing table to neighbors
- Each node **increments** and **appends** its sequence number when sending its local routing table
- When a node decides that a route is **broken**:
 - it increments the sequence number of the route and advertises it with **infinite metric**
 - Destination advertises new sequence number

DSDV: Sequence number setting Rule

Just before node N's broadcast advertisement:

- Node N sets:

$$\text{Seq}(N) \leftarrow \text{Seq}(N) + 2$$

Node N thinks neighbor P is no longer directly reachable

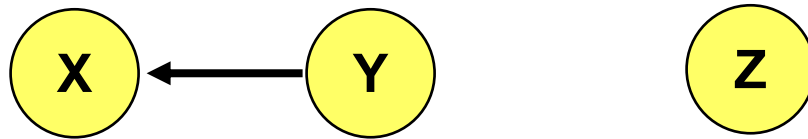
- Node N sets:

$$\text{Seq}(P) \leftarrow \text{Seq}(P) + 1$$

$$\text{Metric}(P) \leftarrow \infty$$

DSDV example

- When X receives information from Y about a route to Z
 - Let the destination sequence number for Z at X be $S(X)$, $S(Y)$ is sent from Y



- If $S(X) > S(Y)$, then X ignores the routing information received from Y (since has more up to date information)
- If $S(X) = S(Y)$, and cost of going through Y is smaller than the route known to X, then X sets Y as the next hop to Z
- If $S(X) < S(Y)$, then X sets Y as the next hop to Z, and $S(X)$ is updated to equal $S(Y)$

Optimized Link State Routing (OLSR)

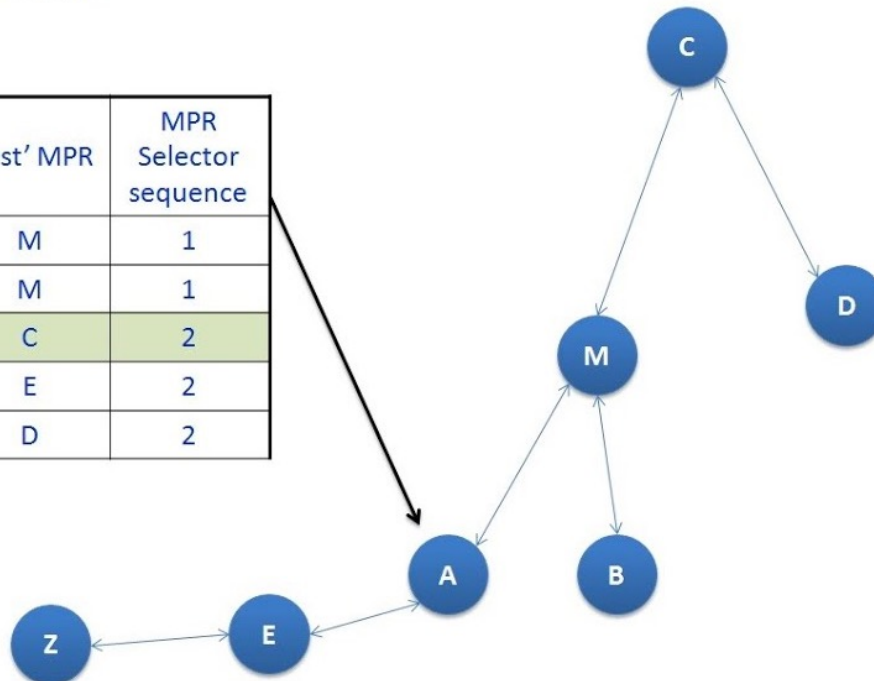
- Links state routing in general: Each node ...
 - ... periodically floods status of its links
 - ... re-broadcasts link state information received from its neighbor
 - ... keeps track of link state information received from other nodes
 - ... uses this information to determine next hop to destinations
- Link state routing is proactive
- Local information is disseminated network wide
- Reduction of overhead of flooding link state information if fewer nodes to forward the information
 - A broadcast from X is only forwarded by selected nodes, so-called **multipoint relays (MPR)**
 - MPRs should globally optimize flooding by optimizing it locally

Optimized Link State Routing (OLSR)

- The Optimized Link-State Routing protocol is characterized by three main functions:
 - Neighbor sensing
 - Optimized flooding (Multi Point Relaying)
 - Link-State messaging and route calculation

A wants to send to D.

Dest' address	Dest' MPR	MPR Selector sequence
B	M	1
C	M	1
D	C	2
Z	E	2
N	D	2



Neighbor sensing

- Each node periodically broadcasts its HELLO messages:
 - Containing the information about **its neighbors** and their **link status**
 - Hello messages are received by all one-hop neighbors
- HELLO message contains:
 - List of addresses of the neighbors to which there exists a valid bi-directional link
 - List of addresses of the neighbors which are heard by node (a HELLO has been received)
 - But link is not yet validated as bi-directional

Neighbor sensing (cont.)

- HELLO messages :
 - Serves **Link sensing**
 - Permit each node to learn the knowledge of its neighbors up to two-hops (**neighbor detection**)
 - On the basis of this information, each node performs the selection of its multipoint relays (**MPR selection signaling**)
 - Indicate selected multipoint relays
- On the reception of HELLO message:
 - Each node constructs its MPR Selector table

Multi-Point Relaying

- Flooding the network with HELLO messages incurs too much overhead
 - OLSR uses multi-point relay (MPR) nodes to decrease the number of unnecessary broadcasts (only selected nodes broadcast HELLO)
- Reduce the number of duplicate retransmissions while forwarding a broadcast packet.
- Restricts the set of nodes retransmitting a packet from all nodes (regular flooding) to a subset of all nodes.
- The size of this subset depends on the topology of the network.

Selectiong Multipoint Relays

MPR Selection

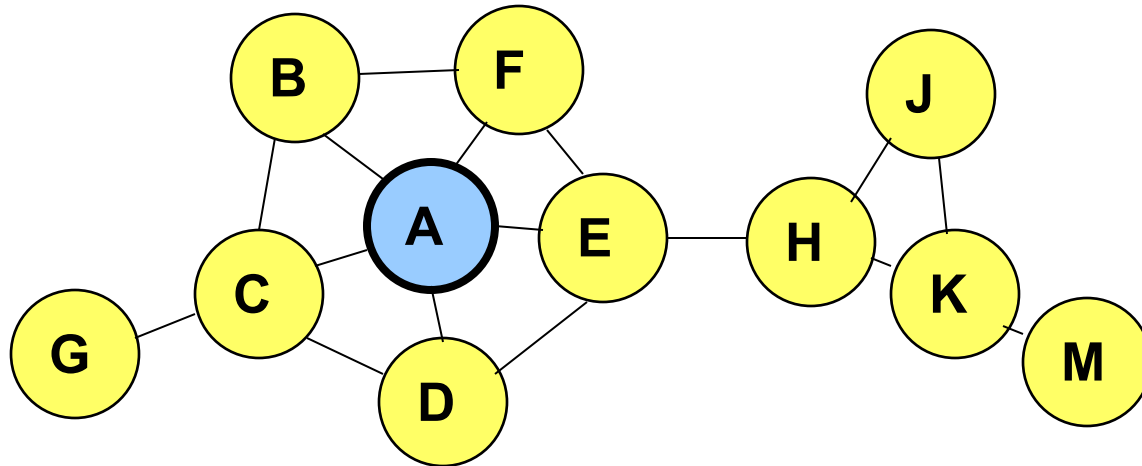
- A node is selected with two rules
 - Any 2-hop neighbor must be covered by at least one multipoint relay
 - The number of multipoint relays should be minimized (per node)
- A node forwards the flooding packets with the following rules
 - Forward if the packet has not already been received
 - The node is multipoint relay of the last emitter

A simple heuristic for computing MPRs

- Start with an empty MPR set
- Add to the MPR set each neighbor that is the only one covering some 2-hop neighbor (must be an MPR anyway)
- Until all 2-hop neighbors are covered repeat:
 - Add to the MPR set a neighbor that covers a maximum of still uncovered 2-hop neighbors

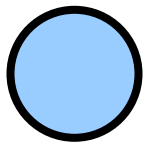
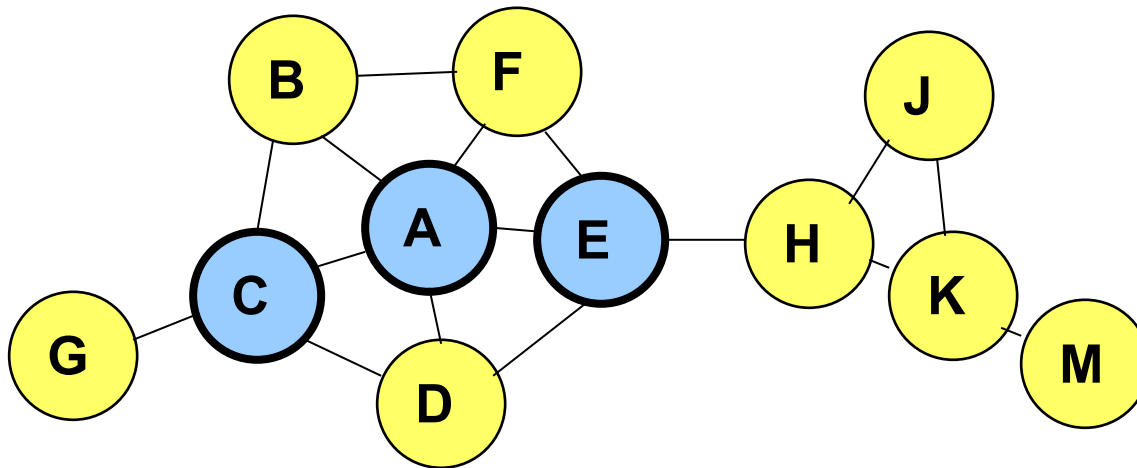
OLSR – MPR Selection (1)

- What are MPRs of node A ?



OLSR – MPR Selection (2)

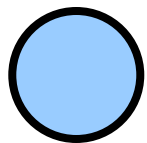
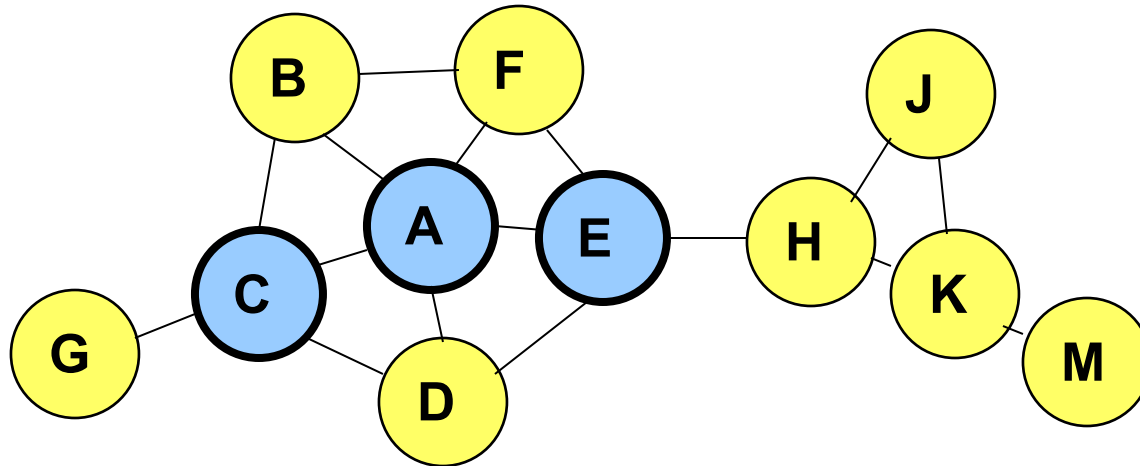
- Nodes C and E forward information received from A
 - What are MPRs of node E?



Node that has broadcast state information from A

OLSR – MPR Selection (3)

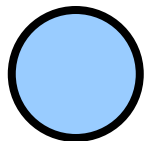
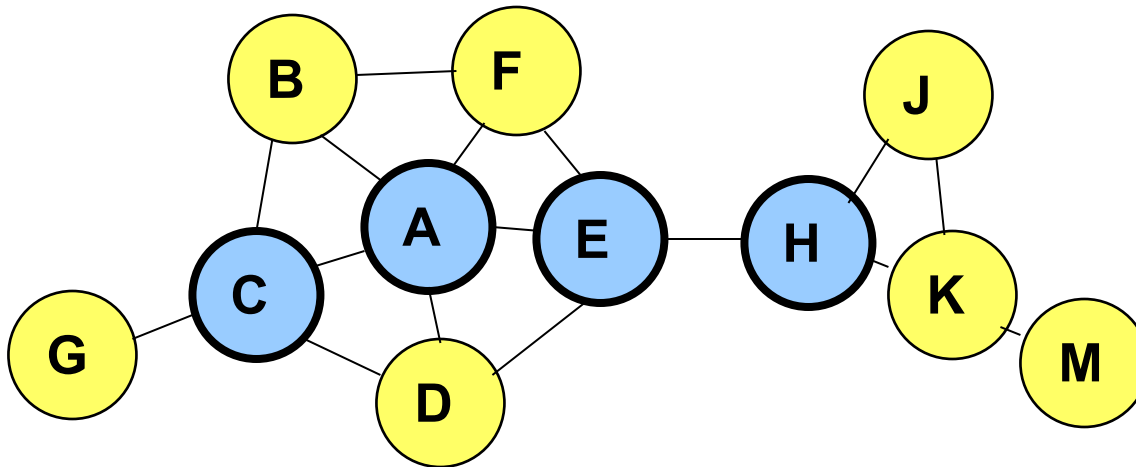
- Nodes A and H are multipoint relays for node E
 - But A has forwarded the link state message already



Node that has broadcast state information from A

OLSR – MPR Selection (4)

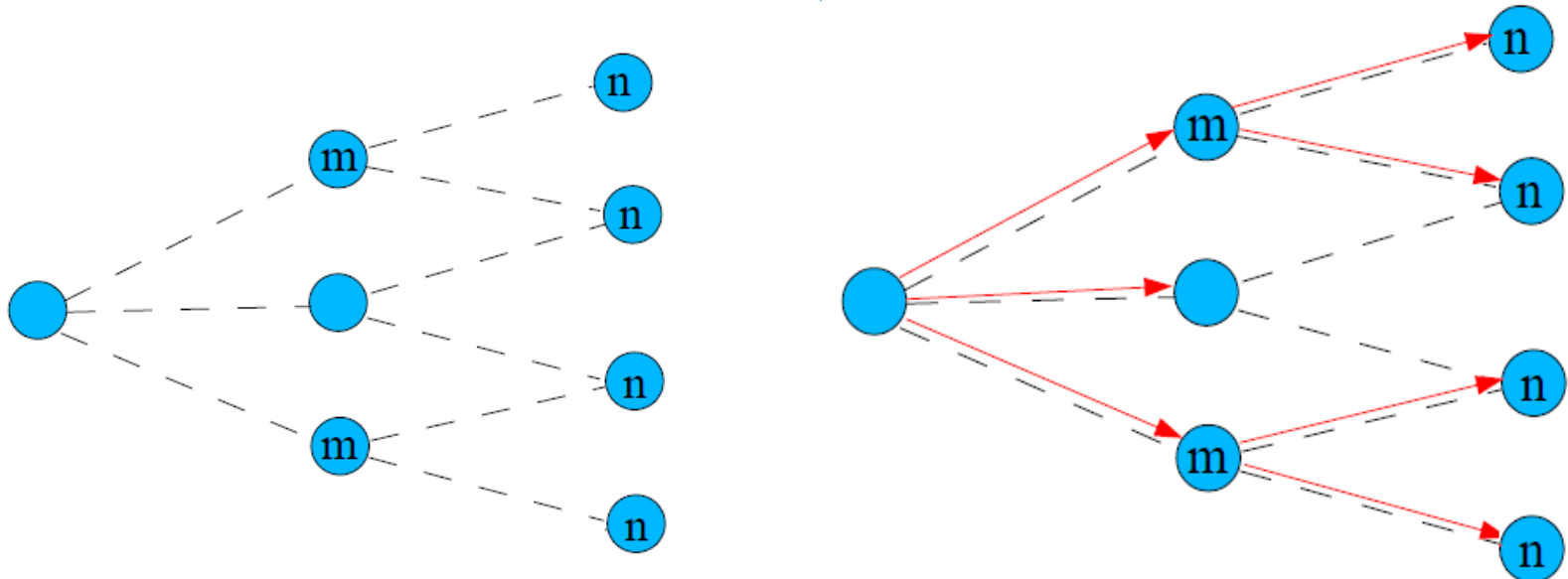
- Node E and K are multipoint relays for node H
 - Node K forwards information received from H in the next step
 - E has already forwarded the same information once



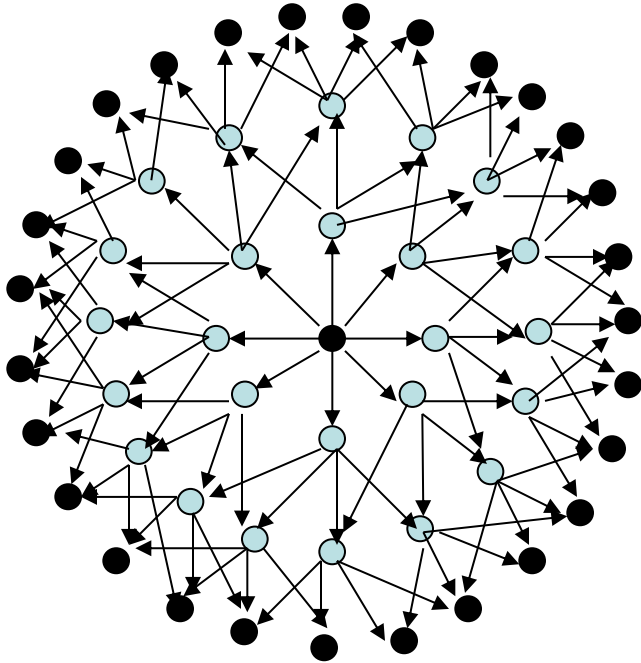
Node that has broadcast state information from A

Link State functionality

- In a classic link-state scheme all nodes flood the network with link-state information.
- OLSR has two link-state optimizations:
 - Only MPR selectors are declared in link-state messages. This minimizes the size of link-state messages.
 - Only nodes selected as MPRs will generate link-state messages. This minimizes the set of nodes emitting link-state messages.

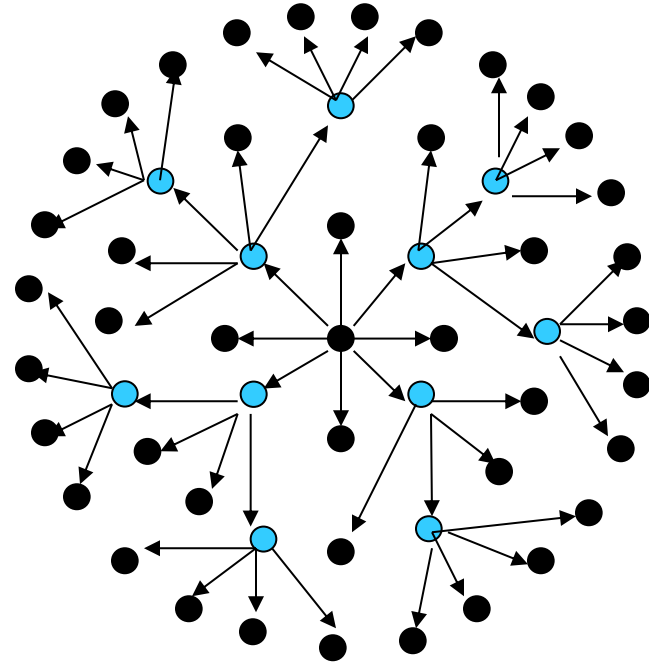


Benefits of Multi-point Relaying



24 retransmissions to diffuse
a message up to 3 hops

● Retransmission node



11 retransmission to diffuse a
message up to 3 hops

● Retransmission node

Reactive Routing Protocols Examples

Source Routing-based: DSR
Backward Learning-based: AODV

Dynamic Source Routing (DSR)

- A **pure reactive** approach relying on source routing
- **Uni-directional** and **asymmetric** routes support
 - (e.g. send a packet to a node D through a route and receive a packet D from another route)
- Excellent performance for routing in multi-hop wireless ad hoc networks
- Very low routing overhead even with continuous rapid motion, which scales to :
 1. zero when nodes are stationary
 2. the affected routes when nodes are moving
- Completely self-organized & self-configuring network
- Entirely **on-demand** operation
- No periodic activity of any kind at any level

Dynamic Source Routing (DSR)

DSR is characterized by three main functionalities:

- **Route Discovery:**

- Initiated by source node S that wants to send packet to destination node D
- **Route Request (RREQ) messages** flood through the net
- Each node *appends own identifier* when forwarding RREQ

- **Route Reply:**

- D on receiving the first RREQ, sends a **Route Reply (RREP)**
- RREP is sent on a route obtained by **reversing** the route appended to received RREQ
- RREP **includes the route** from S to D on which RREQ was received by node D

- **Data Forwarding :**

- S sends data to D by **source routing** through intermediate nodes

Route maintenance in DSR

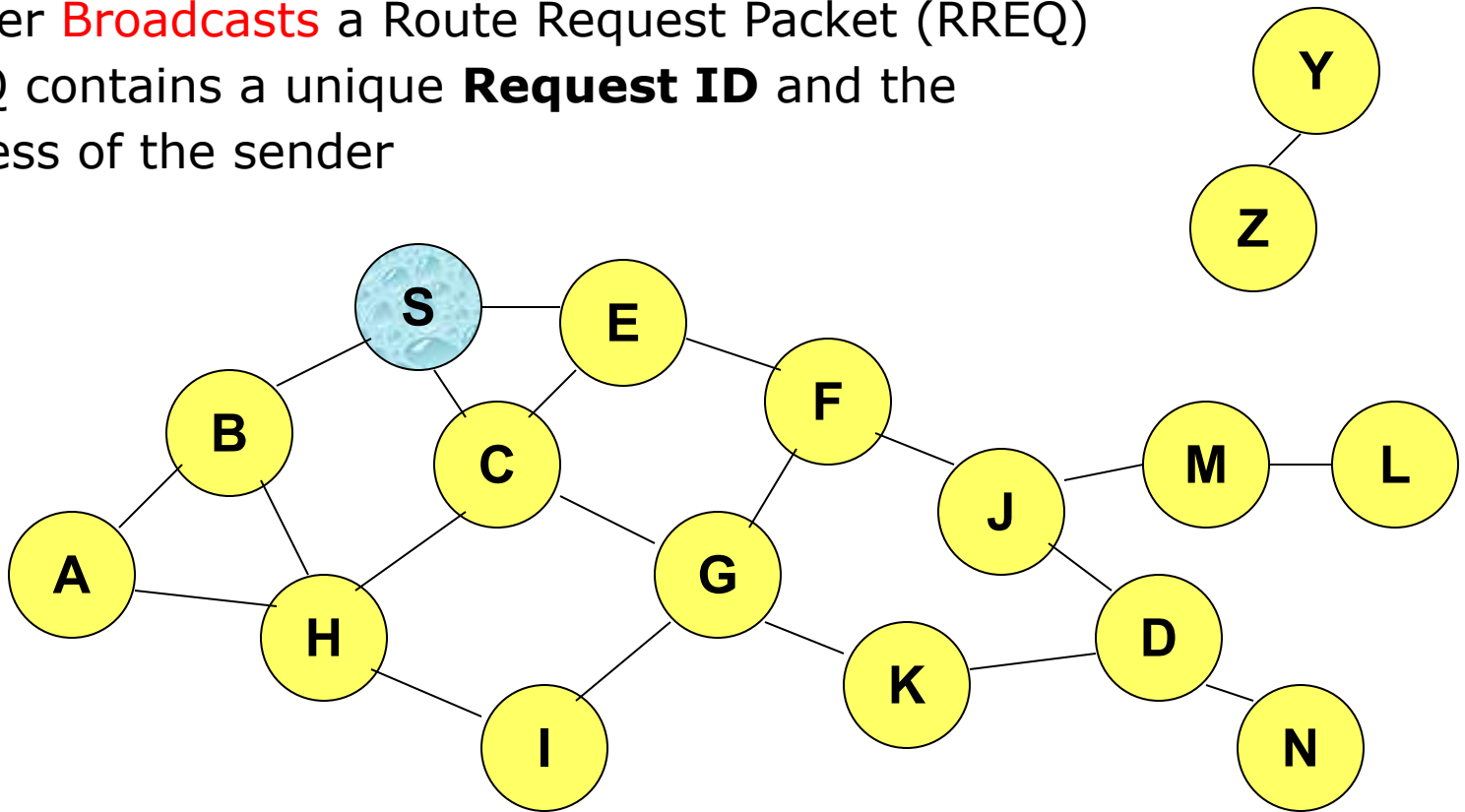
Each node transmitting a packet:

- is responsible for confirming that the packet has been received by the next hop along the source route
- The confirmation it is done with a standard part of MAC layer (e.g. Link-level ACKs)
- If none exists, a DSR-specific software takes the responsibility to sent back an ACK

When retransmissions of a packet in a node reach a maximum number, a Route Error Packet (RERR) is sent from the node back to the source, identifying the broken link

Route discovery in DSR

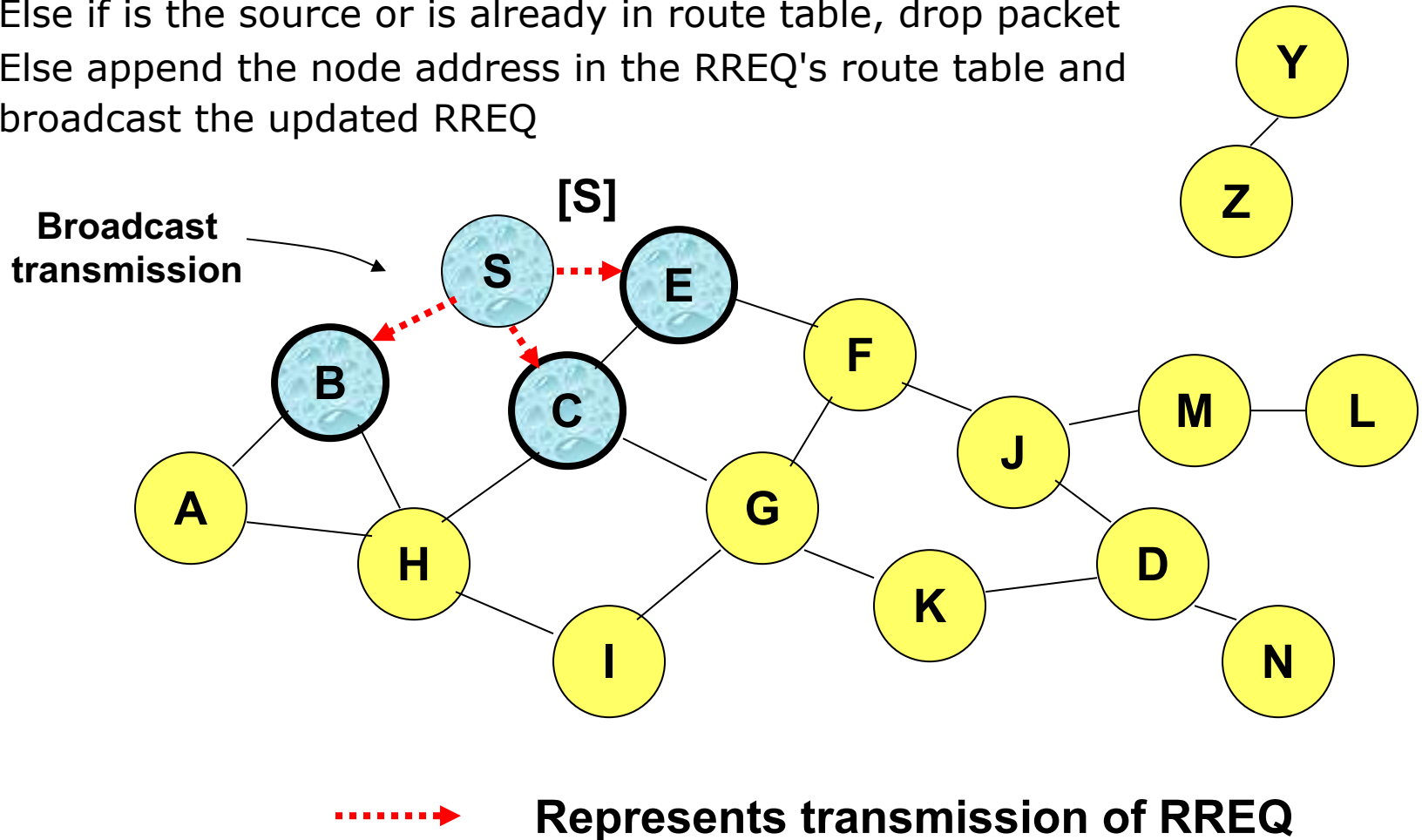
- Sender **Broadcasts** a Route Request Packet (RREQ)
- RREQ contains a unique **Request ID** and the address of the sender



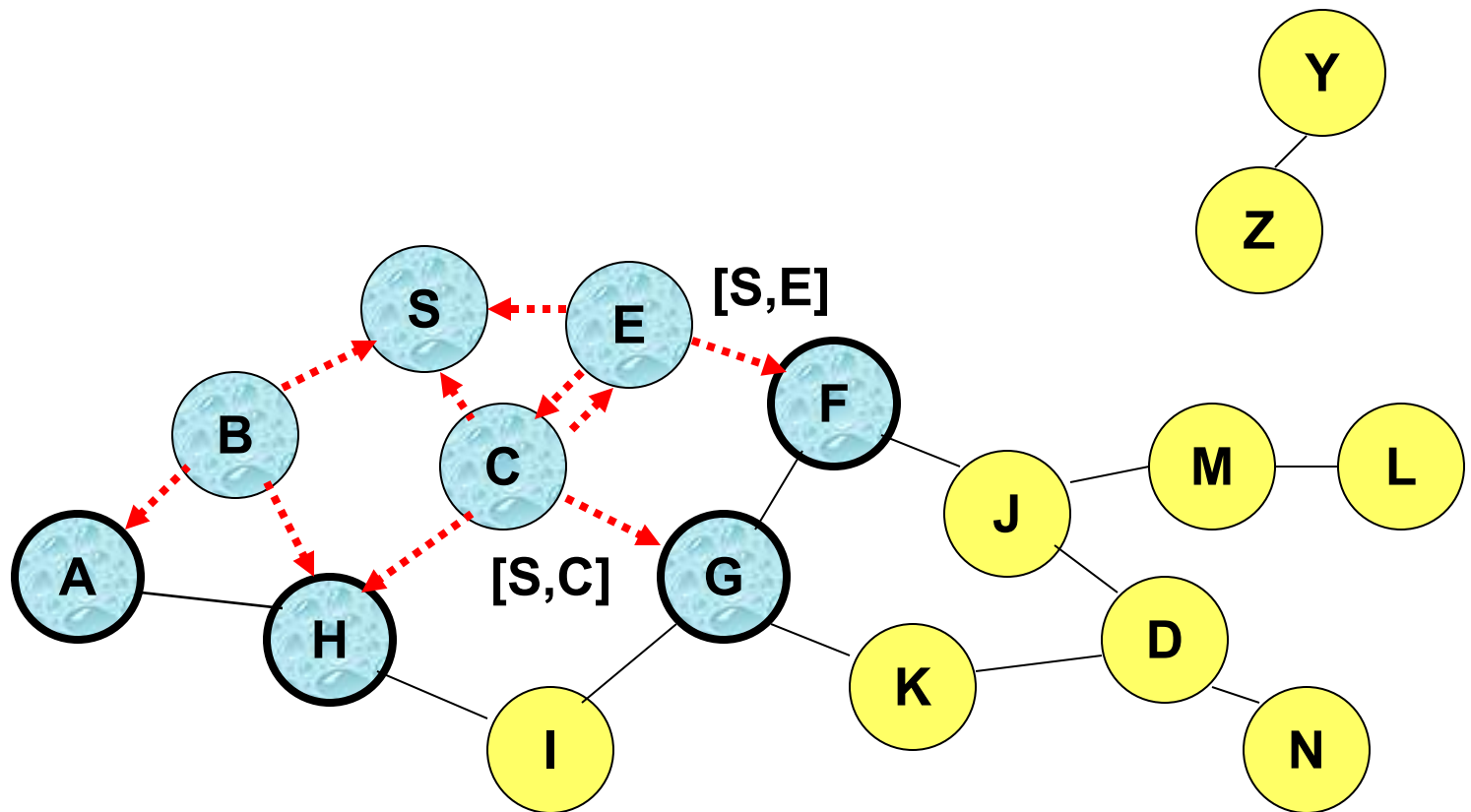
Represents a node that has received RREQ for D from S

Route discovery in DSR

- If this node is the destination, or has route to destination send a **unicast** Route Reply packet (RREP)
- Else if is the source or is already in route table, drop packet
- Else append the node address in the RREQ's route table and broadcast the updated RREQ

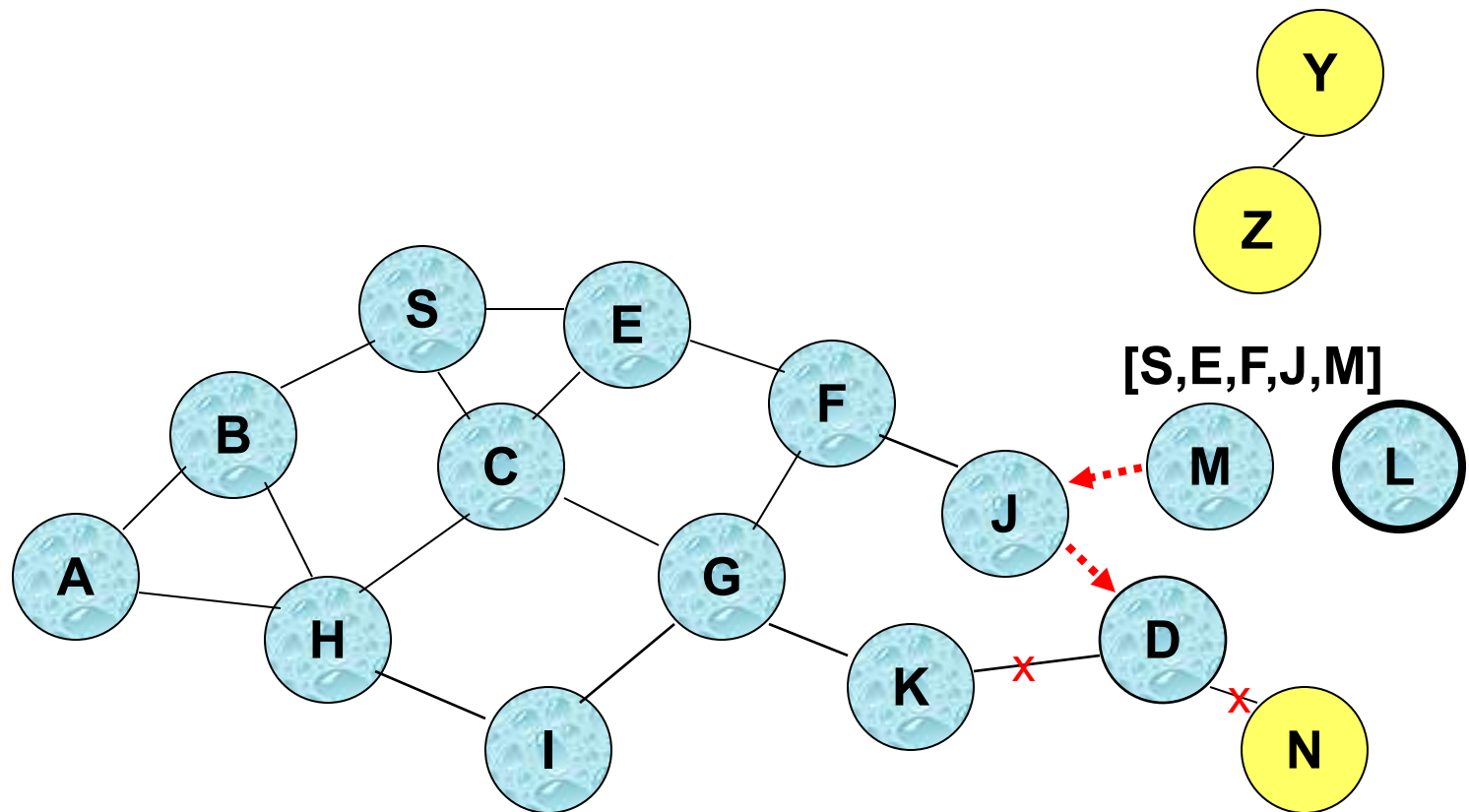


Route discovery in DSR



- Node H receives packet RREQ from two neighbors:
potential for collision

Route discovery in DSR

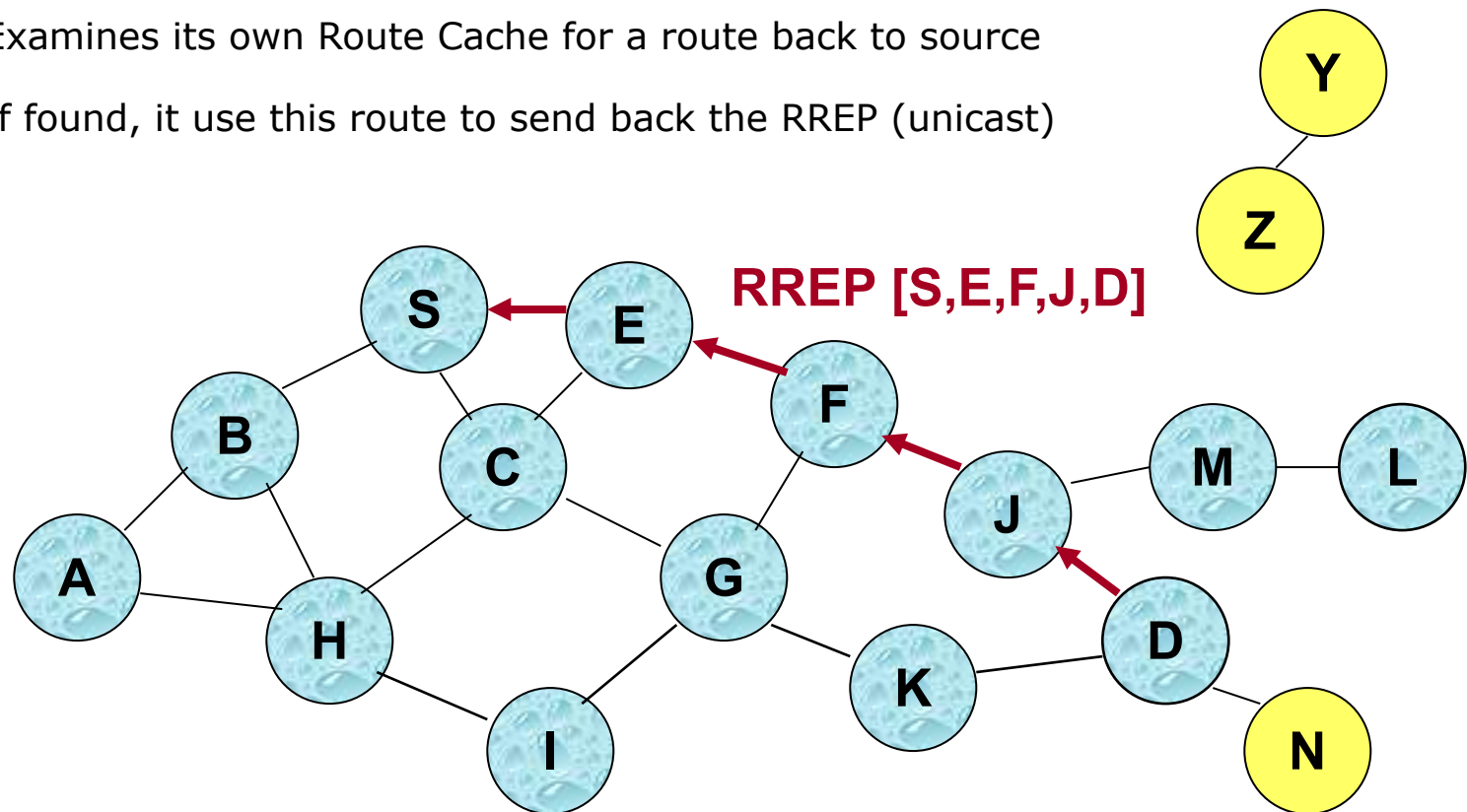


- Node D **does not forward** RREQ, because node D is the **intended target** of the route discovery

Route reply in DSR

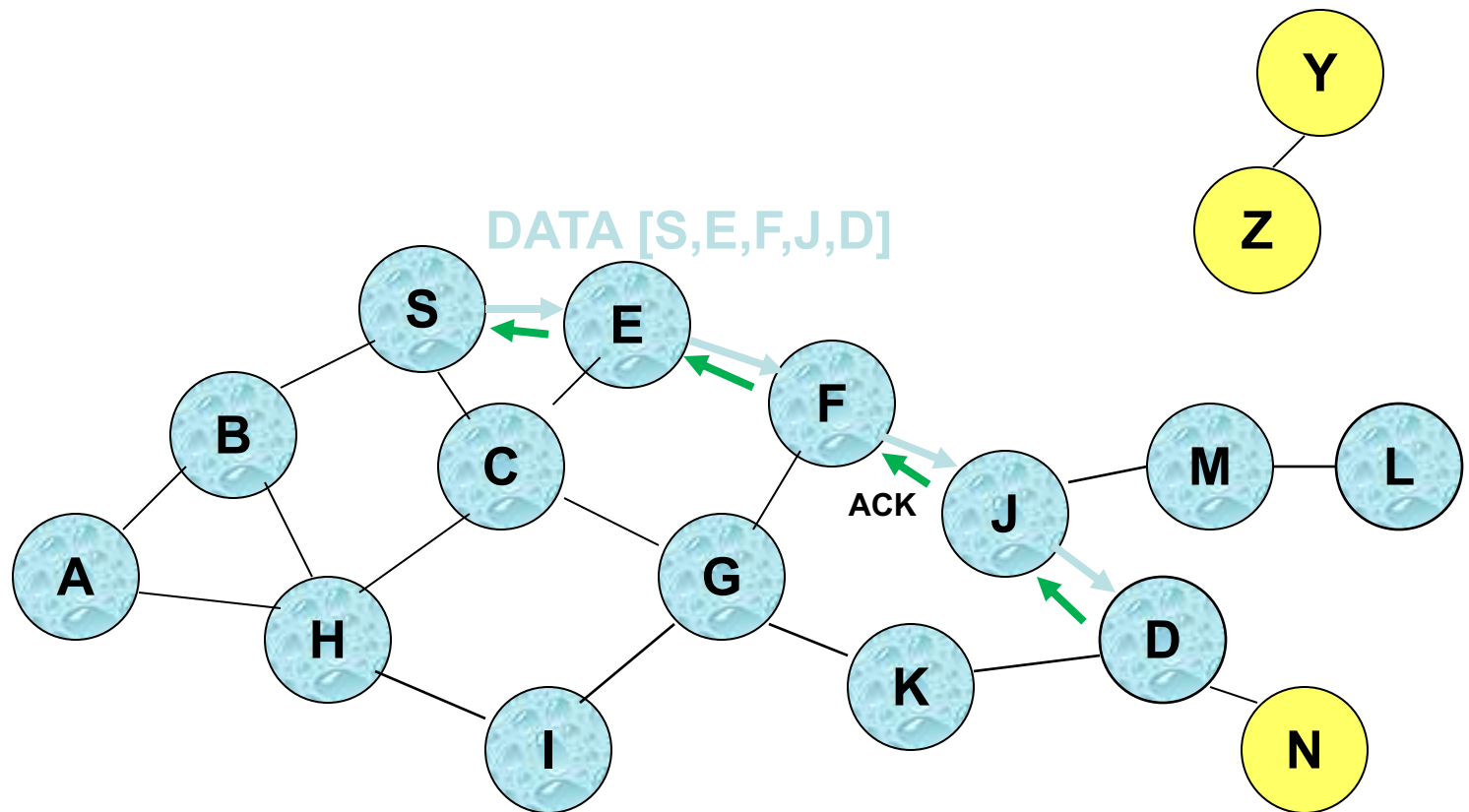
When a RREQ reaches the destination, a RREP must be sent back to source
The destination node:

- Examines its own Route Cache for a route back to source
- If found, it use this route to send back the RREP (unicast)



← Represents RREP control message

Data delivery in DSR



- Packet header size grows with route length due to source routing
- Packets should be acknowledged and after a timeout a RERR message is issued back to the source

Ad Hoc On-demand Distance Vector (AODV)

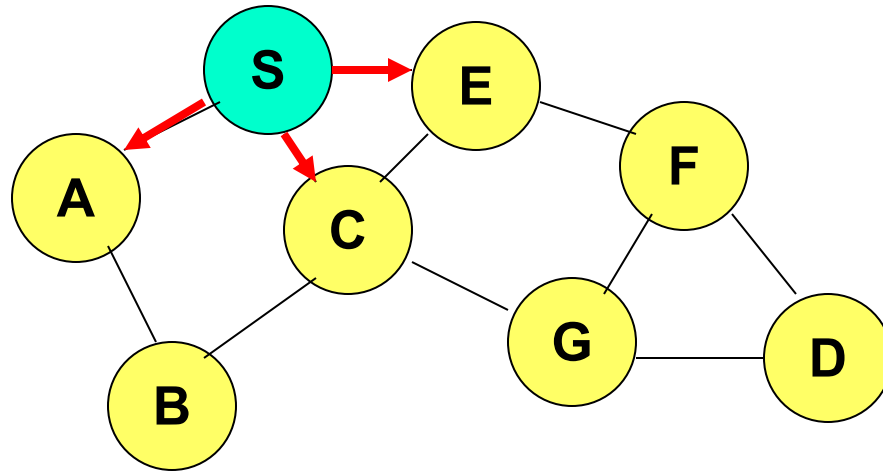
An improved reactive routing protocol:

- Nodes store only the routes they need
- Need for broadcast is minimized
- Reduce memory requirements and needless duplications
- Quick response to link breakage in active routes
- Loop-free routes maintained by use of destination sequence numbers
- Scalable to large populations of nodes

Ad Hoc On-demand Distance Vector (AODV)

- Also AODV creates routes on-demand and uses **symmetric** links between neighboring nodes
- Each node can become aware of its neighborhood by using local broadcasts known as hello messages
 - Nodes learn of their neighbors in one or two ways:
 1. Whenever a node receives a broadcast from a neighbor it updates its local connectivity information about this neighbor
 2. If a neighbor has not sent any packets within an interval it broadcasts a hello message, containing its identity and sequence number
- Neighbor routing tables organized to :
 - optimize response time to local movements
 - provide quick response time for new routes requests

AODV Routing Protocol

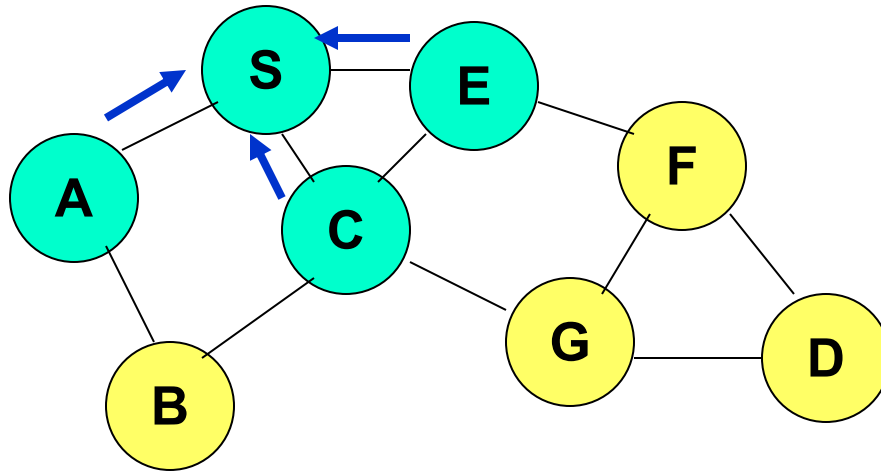


- Source floods route request (RREQ) in the network
- Every node maintains two counters:
 1. node_sequence_number
 2. broadcast_id (number of retries)
- Reverse paths are formed when a node hears a route request.
- Each node forwards the request only once (pure flooding) by checking the sequence number of already forwarded RREQs.

AODV: Use of Sequence Numbers

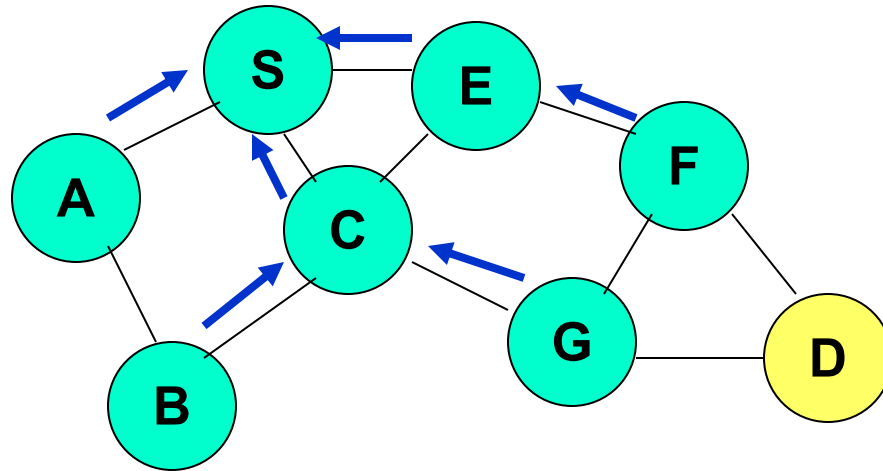
- Each node X maintains a sequence number
 - acts as a time stamp
 - incremented every time X sends any message)
- Each route to X (at any node Y) also has X's sequence number associated with it, which is Y's latest knowledge of X's sequence number.
- Sequence number signifies 'freshness' of the route – higher the number, more up to date is the route.

AODV Route Discovery



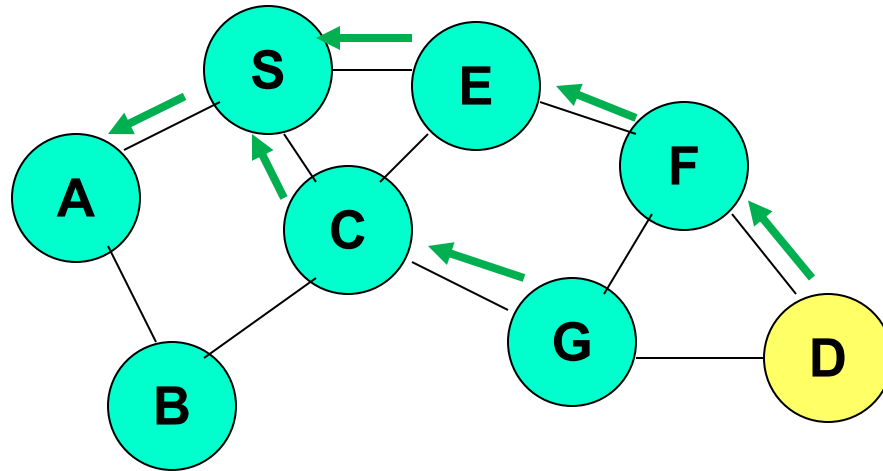
- RREQ structure
 - `<src_addr, src_seq_#, broadcast_id, dest_addr, dest_sequence_#, hop_cnt>`
- **src_addr** and **broadcast_id** uniquely identifies a RREQ
- broadcast_id is incremented whenever source node issues a new RREQ
- Each neighbor either satisfy the RREQ, by sending back a routing reply (RREP), or rebroadcast the RREQ to its own neighbors after increasing the hop_count by one.

AODV Route Discovery



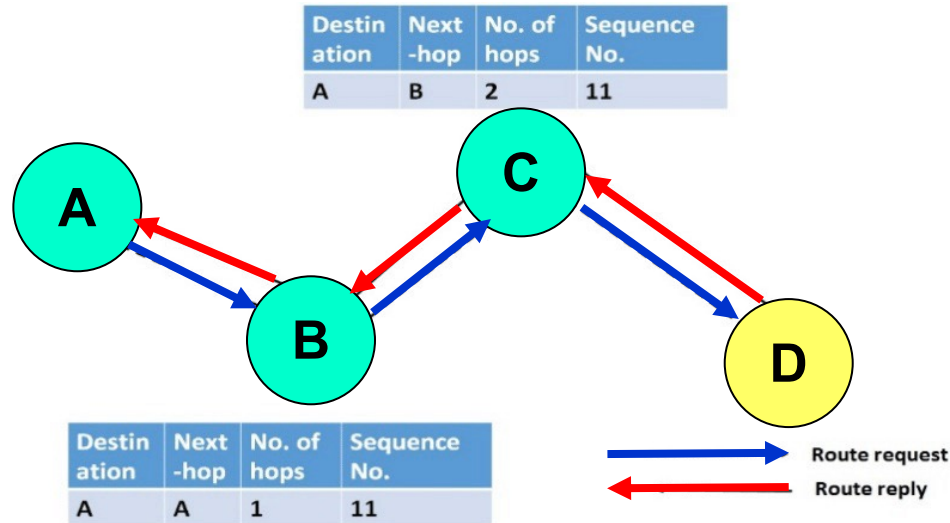
- As RREQ travels from source to many destinations, it sets up the reverse path, from all nodes back to the source.
- Each node records the address of the neighbor from which it received the first copy of the RREQ
- These entries are maintained for at least enough time, for the RREQ to traverse the network and produce a reply (RREP)

AODV Route Discovery



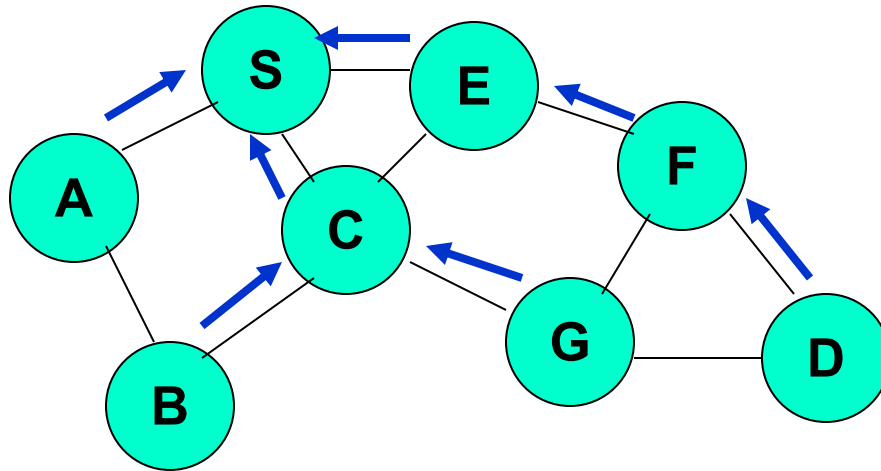
- A route can be determined when the RREQ reaches either the destination D, or an intermediate node G with a 'fresh enough' route to the destination.
- A 'fresh enough' route is a valid route entry for the destination whose associated sequence number is at least as great as that contained in the RREQ.

AODV Route Discovery



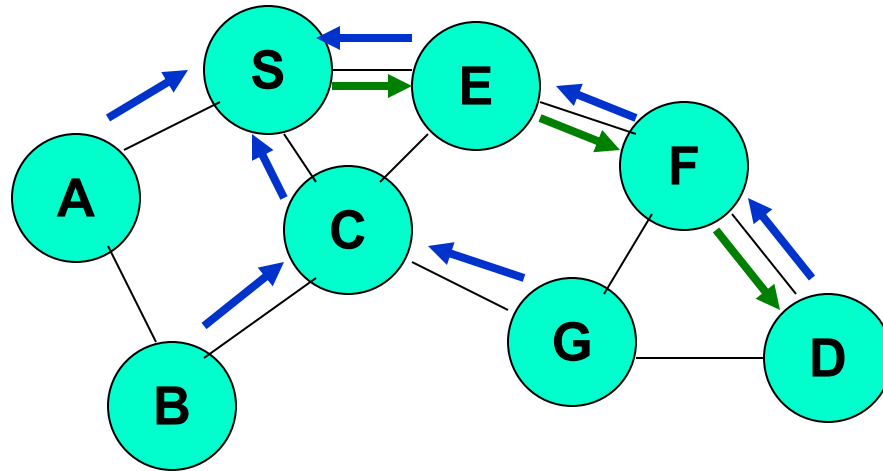
- The AODV routing table are built according to a backward learning scheme based on RREQ messages in order to provide origin visibility according to a backward learning scheme
- Analogously a new route based on destination is set up with RREP messages by populating the routing tables with info related on how to reach the destination
- This information is cached back on all the intermediate nodes and can be used by them for responding to future RREQs

AODV Route Discovery



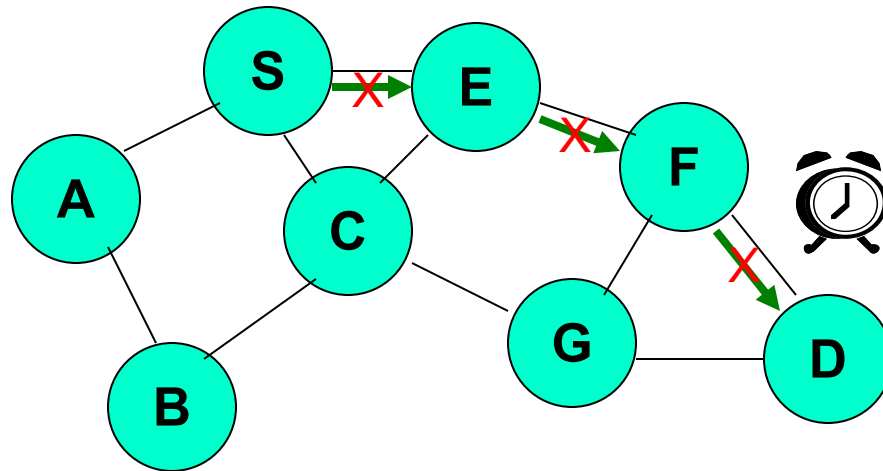
- A node receiving a RREP propagates the first RREP for a given source towards the source (using the reverse path that has already established)
- Nodes that are not in the path determined by the RREP will time out after 3000 ms and will delete the reverse pointers

AODV Route Discovery



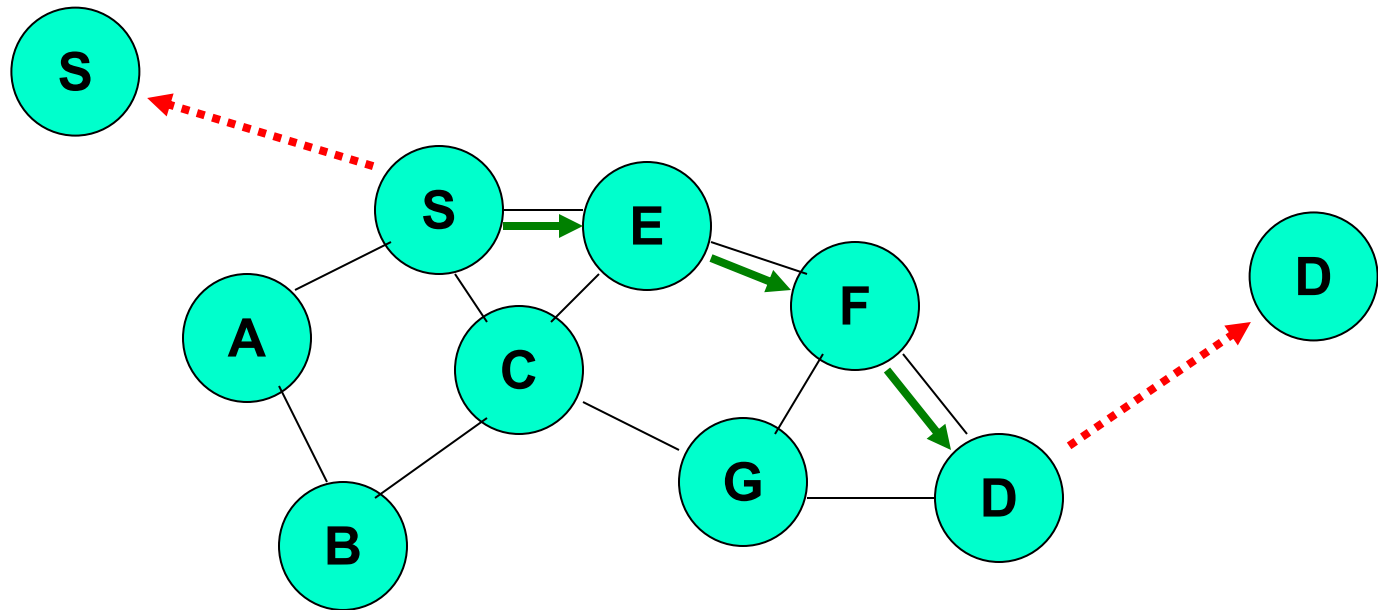
- The number of RREPs towards source is minimized
- The source can begin data transmission as soon as the first RREP received and update later its routing information if it learns of a better route

Route Expiry



- Unused paths expire based on a timer
- Nodes monitor the link status of next hops in active routes. When a link break in an active route is detected, a RERR message is used to notify other nodes that the loss of that link has occurred and the involved destinations are no longer reachable through the broken link.

Effects of Mobility

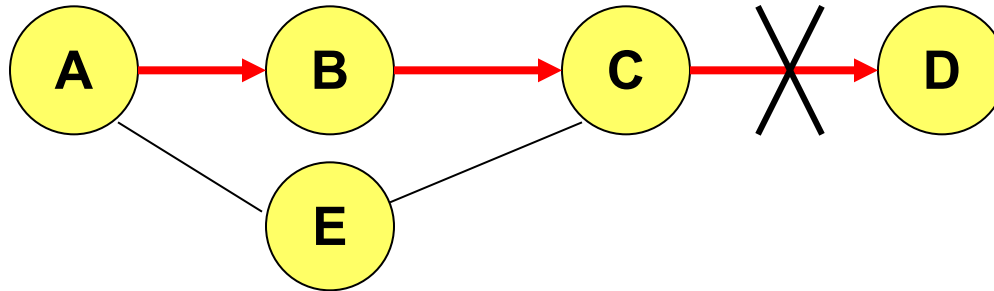


- Movement of nodes not lying along an active path does NOT affect the route to that path's destination
- If the source node moves, it can simply re-initiate the route discovery procedure
- If the destination or some intermediate node moves, a special RREP is sent to the affected nodes
- When a node is unreachable a special RREP is sent back towards the source, containing a new sequence number and hop count of ∞

AODV – RREP Optimization

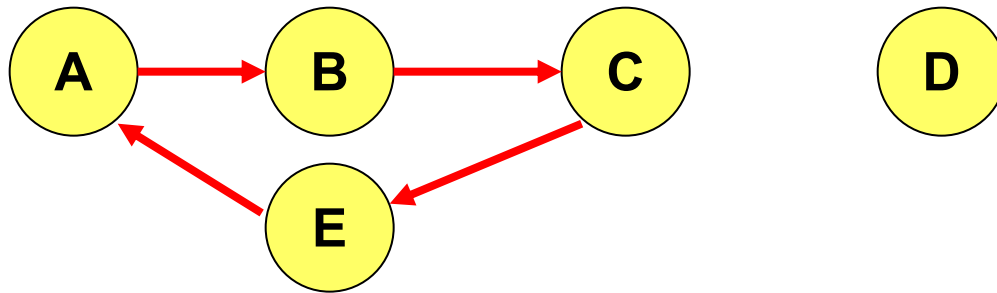
- An intermediate node with a route to D already in place can reply to route request
 - Faster operation.
 - Quenches route request flood.
- Such optimization can cause routing loops in presence of link failures

AODV: Routing Loops



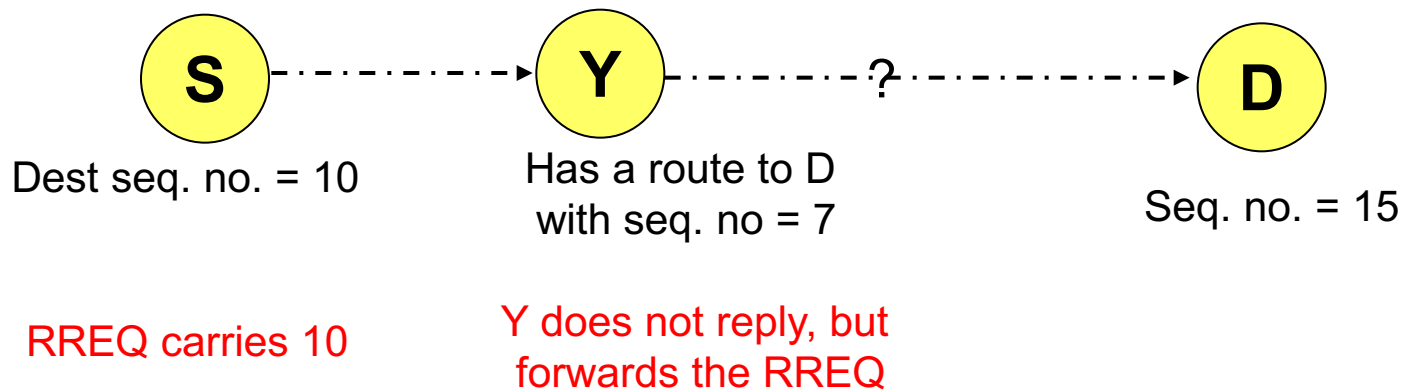
- Assume, link C-D fails, and node A does not know about it (route error packet from C is lost).
- C performs a route discovery for D.
- Node A receives the route request (via path C-E-A)
- Node A replies, since A knows a route to D via node B

AODV: Routing Loops



This Results in a loop!: C-E-A-B-C

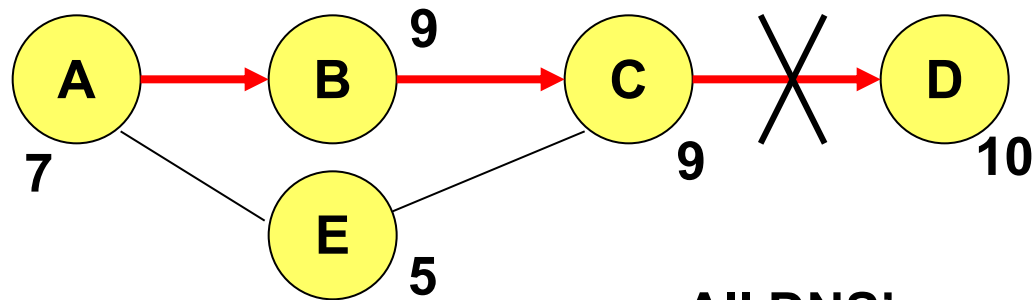
Use of Sequence Numbers in AODV



- Loop freedom: Intermediate node replies with a route (instead of forwarding request) only if it has a route with a higher associated sequence number.

Avoidance of Loop

DSN = Destination Sequence Number.



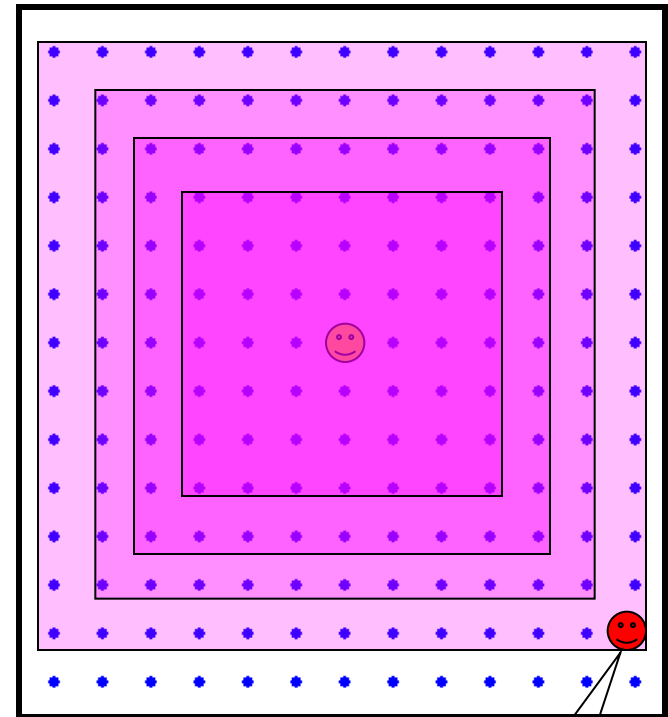
All DNS's are for D

- Link failure increments the DSN at C (now is 10).
- If C needs route to D, RREQ carries the DSN (10).
- A does not reply as its own DSN is less than 10.

Expanding Ring Search

Another useful optimization reducing overhead in route discovery

- Enforce Hop Limit on RREQs
- Use TTL in the packet header to specify the first ring boundary
- RREQ is initially forwarded n times (n hops)
- If destination is not within n -hop
 - Increase TTL to a larger value
 - Restart discovery
 - Reiterate the process a number of times before giving up

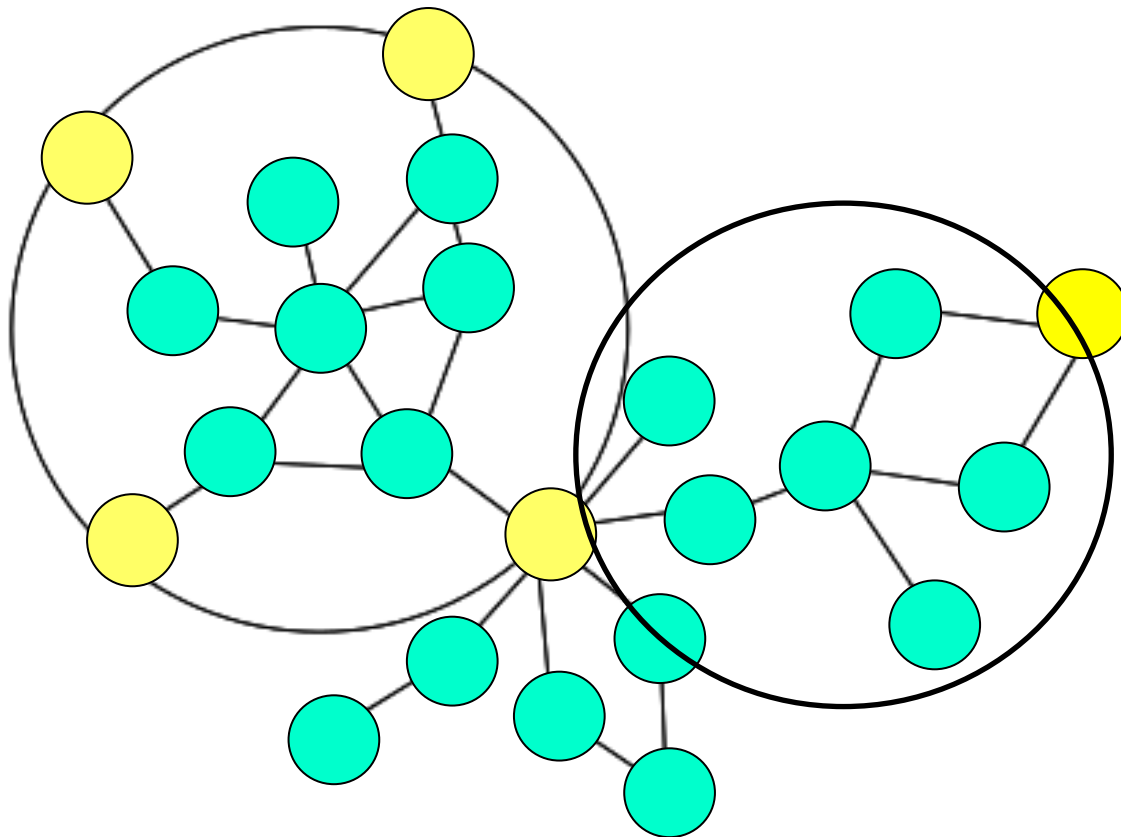


This is useful if destination is close to the source

Hi!

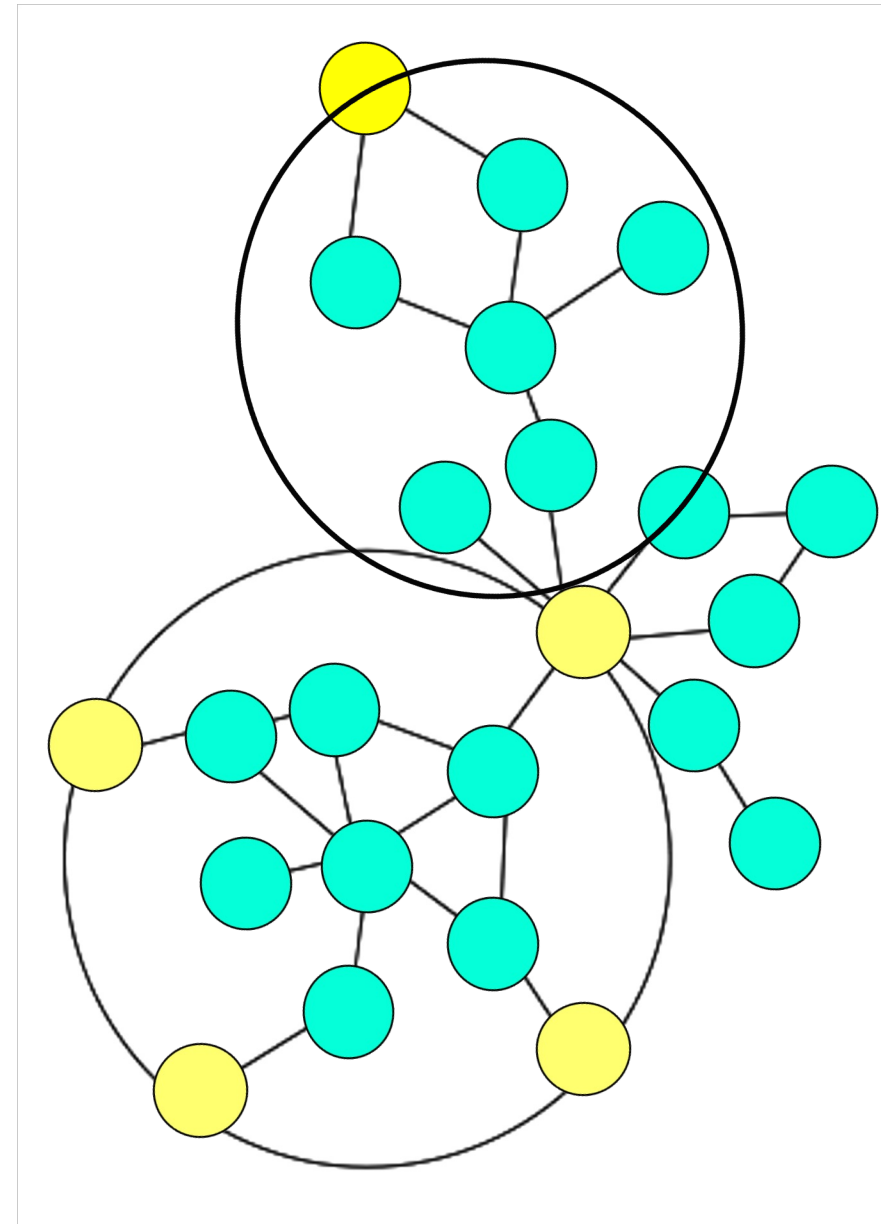
The Zone Routing Protocol (ZRP)

- The First hybrid routing protocol combining the best properties of proactive and reactive routing protocols
 - Reduce control overhead of proactive routing protocols
 - Decrease latency caused by route search operations in reactive approaches



The Zone Routing Protocol (ZRP)

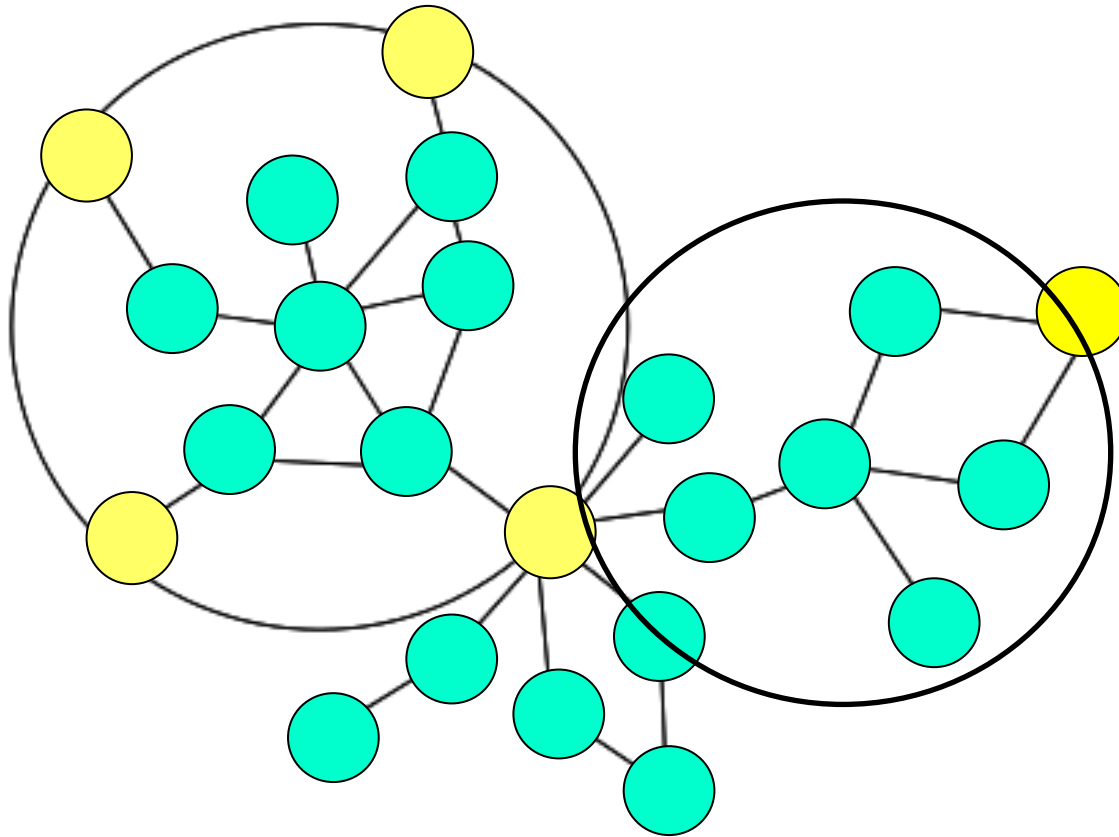
- Network is divided into routing zones with a radius d in hops.
- The zone thus includes the nodes, whose distance from the node in question is at most d hops.
- Given a node N :
 - nodes within hop distance at most $d \Rightarrow N$'s routing zone.
 - Peripheral nodes: nodes which are exactly d hops away from N .
- The number of nodes in the routing zone can be regulated by adjusting the transmission power
 - Lowering the power reduces the number of nodes within direct reach and vice versa.



The Zone Routing Protocol (ZRP)

- Different routing approaches are exploited.
- **Inter-zone** : Inter-zone Routing Protocol (IERP)
 - reactive routing protocols that offer enhanced route discovery and route maintenance services
- **Intra-zone** : Intra-zone Routing Protocol (IARP)
 - limited-depth, proactive link-state routing protocols.
 - maintains routing information for nodes that are within the routing zone of the node.
- Instead of broadcasting packets, ZRP uses a concept called bordercasting.
 - utilizes the topology information provided by IARP to direct query request to the border of the zone.

The Zone Routing Protocol (ZRP)



- In order to detect new neighbor nodes ZRP relies on a Neighbor Discovery Protocol (NDP) provided at the Media Access Control layer.
- NDP transmits “HELLO” beacons at regular intervals. Upon receiving a beacon, the neighbor table is updated.

The Zone Routing Protocol (ZRP)

- If a packet's destination is in the same zone as the origin, the proactive protocol using an already stored routing table is used to deliver the packet immediately
- For routes outside the local zone, discovery is reactive.
 - Source node sends a route request to the border nodes of its zone, containing its own address, the destination address and a unique sequence number to check each successive zone in the route to see whether the destination is inside that zone.
 - Each border node checks its local zone for the destination.
 - If the destination is not a member of this local zone, the border node adds its own address to the route request packet and forwards the packet to its own border nodes.
 - If the destination is a member of the local zone, it sends a route reply on the reverse path back to the source.
 - The source node uses the path saved in the route reply packet to send data packets to the destination.

The Zone Routing Protocol (ZRP)

- The reactive routing process is divided into two phases:
 - route request
 - route reply
- In the route request, the source sends a route request packet to its peripheral nodes using bordercasting
- If the receiver of a RREQ packet knows the destination, it responds by sending a RREP back to the source.
- Otherwise, it continues the process by bordercasting the packet.
- The reply is sent by any node providing a route to destination.
- The source can receive the complete source route to the destination. Alternatively, the nodes along the path to the destination record the next-hop address in their routing table.

